HITCHIN'S α-INVARIANT AND POSITIVE SECTIONAL CURVATURE ON HOMOTOPY 9-SPHERES

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Abstract. We prove that a smooth homotopy 9-sphere admits a Riemannian metric of positive sectional curvature if and only if its Hitchin α-invariant vanishes, or equivalently if and only if it admits a Riemannian metric of positive scalar curvature.

1. Introduction

For a closed spin n-manifold M, Hitchin's α-invariant is the real index of the spin Dirac operator, taking values in the real K-homology group of a point:

$$\alpha(M) \in KO_n(\mathrm{pt}).$$

Hitchin proved that this real Dirac index vanishes whenever M admits a metric of positive scalar curvature [10]. Conversely, Stolz proved that its vanishing is sufficient when M is also simply connected and has dimension at least five [19].

Positive sectional curvature implies positive scalar curvature. For smooth homotopy 9-spheres, we prove that the converse holds at the level of existence of metrics. A smooth homotopy 9-sphere is a closed smooth manifold homotopy equivalent to S^9; it is *exotic* if it is not diffeomorphic to the standard sphere. Every homotopy 9-sphere is orientable; after an orientation is fixed, its spin structure is unique. Dimension nine is the first dimension in which a simply connected homotopy sphere can have nonzero α-invariant [4, Section 2.3].

For a Riemannian metric g, write $\langle\cdot,\cdot\rangle_g$ for its tangent-space inner product and let ∇ be its Levi–Civita connection. We use the curvature convention

$$R^g(X,Y)Z = \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z.$$

For linearly independent $X, Y \in T_pM$, the sectional curvature of $\Pi = \mathrm{span}\{X, Y\}$ is

$$\sec_g(\Pi) = \frac{\langle R^g(X,Y)Y, X\rangle_g}{\langle X,X\rangle_g\langle Y,Y\rangle_g - \langle X,Y\rangle_g^2}. \tag{1.1}$$

The notation $\sec_g > 0$ means that this quantity is positive at every point and on every two-plane. We write Scal_g for scalar curvature; for a g-orthonormal frame $(e_1,\dots,e_n)$ it satisfies $\mathrm{Scal}_g = 2\sum_{i<j}\sec_g(\mathrm{span}\{e_i,e_j\})$. All metrics considered below are smooth.

1.1. Main result.

Theorem 1.1. *Let Σ be a smooth homotopy 9-sphere. The following conditions are equivalent:*

(i) *Σ admits a Riemannian metric of positive sectional curvature;*

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(ii) Σ *admits a Riemannian metric of nonnegative sectional curvature;*
(iii) Σ *admits a Riemannian metric of positive scalar curvature;*
(iv) $\alpha(\Sigma) = 0 \in KO_9(\mathrm{pt}) \cong \mathbb{Z}/2$.

To prove (iv)⇒(i), we construct a positively curved metric for each nonstandard smooth type with vanishing index. The standard sphere has its round metric. The implication (iii)⇒(iv) is Hitchin's obstruction. Its extension to nonnegative sectional curvature is proved in proposition 2.1.

To list the smooth types, let Θ_9 be the group of oriented homotopy 9-spheres under connected sum. The classical classification gives

$$\Theta_9 \cong (\mathbb{Z}/2)^3. \tag{1.2}$$

Write $\pi_j^s = \varinjlim_k \pi_{j+k}(S^k)$ for the jth stable homotopy group of spheres, with transition maps given by suspension. Let $SO = \varinjlim_k SO(k)$, where the inclusions add an identity coordinate, and let $J_j : \pi_j(SO) \to \pi_j^s$ be the stable J-homomorphism. We put $\operatorname{Coker} J_j = \pi_j^s / \operatorname{Im} J_j$ and use $[u]$ for the image of $u \in \pi_j^s$ in this quotient. The subgroup bP_{10} consists of homotopy 9-spheres that bound parallelizable 10-manifolds. There is a split exact sequence

$$0 \longrightarrow bP_{10} \longrightarrow \Theta_9 \xrightarrow{p_9} \operatorname{Coker} J_9 \longrightarrow 0, \tag{1.3}$$

with

$$bP_{10} = \mathbb{Z}/2\{\kappa\}, \qquad \operatorname{Coker} J_9 = \mathbb{Z}/2\{[\eta\epsilon]\} \oplus \mathbb{Z}/2\{[\mu]\}. \tag{1.4}$$

Here κ denotes the oriented class of the Kervaire sphere, the boundary of the plumbing of two tangent disc bundles of S^5. We use Toda's notation $\eta \in \pi_1^s$, $\nu \in \pi_3^s$, $\epsilon \in \pi_8^s$, and $\mu \in \pi_9^s$. The relation $[\eta\epsilon] = [\nu^3]$ holds in $\operatorname{Coker} J_9$; the relevant stable homotopy calculations are recalled in Section 2 [11, 20, 21].

Our construction specifies a lift $\lambda \in \Theta_9$ of $[\eta\epsilon]$. Choose $\widehat{\mu} \in \Theta_9$ with $p_9(\widehat{\mu}) = [\mu]$. Then

$$\Theta_9 = \langle\kappa\rangle \oplus \langle\lambda\rangle \oplus \langle\widehat{\mu}\rangle, \qquad \ker\alpha = \{0, \kappa, \lambda, \kappa+\lambda\}. \tag{1.5}$$

The four classes in $\widehat{\mu} + \ker\alpha$ have nonzero index [10, 21, 17]. Table 1 records theorem 1.1 for all eight classes. Orientation reversal does not change these entries, since every element of Θ_9 has order at most two.

Class in Θ_9	Image under p_9	α	Admits positive sectional curvature?
0	0	0	Yes
κ	0	0	Yes
λ	$[\eta\epsilon]$	0	Yes
$\kappa+\lambda$	$[\eta\epsilon]$	0	Yes
$\widehat{\mu}$	$[\mu]$	1	No
$\widehat{\mu}+\kappa$	$[\mu]$	1	No
$\widehat{\mu}+\lambda$	$[\mu]+[\eta\epsilon]$	1	No
$\widehat{\mu}+\kappa+\lambda$	$[\mu]+[\eta\epsilon]$	1	No

TABLE 1. The classification of positive sectional curvature on the eight oriented smooth homotopy 9-spheres.

1.2. **Related work.** Chen, Ni, and Zhang [3] construct a positively curved metric on the Kervaire 9-sphere. Subsequently, Chen, Deng, Ni, and Zhang [2, Theorem 1.2] extended their preprint to obtain the same classification as theorem 1.1.

A complete draft of the present work had already been prepared before that revision appeared. The two approaches share certain topological ingredients, including a four-periodic normal-framing calculation, but differ in their geometric constructions. In particular, the treatment of the remaining nonstandard class with vanishing α-invariant in [2, Sections 10–12] requires a positively curved cylinder, a four-form modification, and two disc fillings. In contrast, our invariant-transition theorem theorem 6.7 provides a direct two-disc construction, yielding a unified geometric proof for all three nonstandard classes in $\ker \alpha$.

The argument here realizes all three nonstandard classes by gluing two quotient discs directly. We identify the Kervaire class through Sperança's pullback model [18], and identify λ using a family defining his exotic 8-sphere. The third attaching map comes from left quaternionic multiplication and an invariant transition function. It is treated by theorem 6.7.

The metric construction uses the two-disc method of He–Liu–Yau [9]. Galaz-García [6] and Deng–Hu–Zhang [5] give versions for conjugation-equivariant transition functions; the latter allows arbitrary orthogonal S^3-representations. These supply the geometric framework for the classes κ and λ. We give the curvature estimates and boundary comparisons for both types of transition in Section 6.

1.3. **Outline of the proof.** We first construct three attaching diffeomorphisms of S^8 and identify the smooth homotopy spheres obtained by gluing two 9-discs along them. The maps are defined by left and right quaternionic multiplication in a tubular neighborhood of a standard $S^3 \subset S^8$. Their formulas, support conditions and normal coordinates are fixed in Section 3.

The first sphere is identified with Sperança's Kervaire model. The second is constructed from a family of disc diffeomorphisms defining his exotic 8-sphere. Precomposition with the Hopf map on the classifying space of the diffeomorphism group gives a 9-sphere with image $[\eta\epsilon]$ under p_9. We denote this class by λ. A relative mapping-class calculation and quaternionic inversion identify the third sphere as $\kappa + \lambda$.

For each attaching map we construct positively curved metrics on the two discs. The southern metric is a quotient of a connection metric, and the northern metric is a quotient of a doubly warped product. For the southern connection metric over (B, g_B), the mixed curvature contains $-r^{-1}\operatorname{Hess}_{g_B} r$, where r is the fiber radius. A strongly concave function $\varphi : B \to \mathbb{R}$ supplies the positive term when $r^2 = \varepsilon e^{\varepsilon\varphi}$; the scale $\varepsilon > 0$ is then chosen small relative to the connection curvature and its covariant derivative. Both quotient actions are free. The estimates apply to every horizontal two-plane, including those over points where the base action has a nontrivial stabilizer.

Let $\phi : \partial D_N \to \partial D_S$ be the attaching map already used to identify the smooth sphere. If h_N, h_S are the induced boundary metrics and B_N, B_S their outward second fundamental forms, the constructed metrics satisfy

$$h_N = \phi^* h_S, \qquad B_N + \phi^* B_S > 0.$$

Here $B(Y, Z) = \langle \nabla_Y n, Z \rangle_g$ for the outward unit normal n, and > 0 means positive definiteness. The Reiser–Wraith theorem [15, Theorem A(i), $k = 1$] then gives a

smooth positively curved metric on the specified sphere. Section 2 fixes the disc parametrizations used in these boundary comparisons.

Sections 4 and 5 identify the three smooth types; Section 6 constructs the disc metrics, and Section 7 completes the proof. The appendices contain the connection-metric estimates, warping functions, and parameterized smoothing comparison.

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2. Homotopy 9-spheres and clutching conventions

2.1. **The group Θ_9.** For the dimensions $n = 8, 9$ used below, let Θ_n denote the group of oriented diffeomorphism classes of smooth homotopy n-spheres, with connected sum as group operation. The clutching construction identifies this group with orientation-preserving diffeomorphisms of S^{n-1} modulo those extending over D^n [11, 1]. We use the convention

$$\Sigma_\varphi = D^n_N \cup_\varphi (-D^n_S) :$$

φ is orientation preserving in the two standard boundary parameters, while the southern disc is given the opposite orientation. With this convention composition represents connected sum. We suppress the orientation sign in subsequent gluing expressions. Reversing the convention negates the sphere class, which leaves it unchanged in Θ_8 and Θ_9.

The Kervaire–Milnor map p_n records a stable normal framing modulo the stable J-image. More precisely, a homotopy n-sphere admits a stable normal framing. The Pontryagin–Thom collapse associated with such a framing represents an element of π^s_n; changing the framing changes this element by an element of $\operatorname{Im} J_n$. Its class in $\operatorname{Coker} J_n$ is $p_n([\Sigma])$ [11]. We use this convention for $n = 8, 9$ and write $p_n(\Sigma)$ for $p_n([\Sigma])$.

The group structure and exact sequence are given in (1.2)–(1.4). We use the stable-stem notation of Toda and Wang; in particular, $\bar\nu \in \pi^s_8$. In degree nine,

$$\pi^s_9 = \mathbb{Z}/2\{\eta\bar\nu\} \oplus \mathbb{Z}/2\{\eta\epsilon\} \oplus \mathbb{Z}/2\{\mu\}, \qquad \operatorname{Im} J_9 = \mathbb{Z}/2\{\eta\bar\nu + \eta\epsilon\}. \tag{2.1}$$

The degree-eight and degree-nine entries of [21, Appendix C, Table 4] give the displayed groups and $\operatorname{Im} J_9$. Relation C.2(1) there is $\eta\bar\nu = \nu^3$; hence $[\eta\epsilon] = [\nu^3]$ in $\operatorname{Coker} J_9$.

In degree eight, $\Theta_8 \cong \mathbb{Z}/2$, and the Kervaire–Milnor map $p_8 : \Theta_8 \to \operatorname{Coker} J_8 \cong \mathbb{Z}/2\{[\epsilon]\}$ is an isomorphism [11, 21]. We write $\Sigma^8_{[\epsilon]}$ for its nonzero class. The two lifts of $[\eta\epsilon]$ differ by κ; Section 4 specifies λ by a geometric construction.

2.2. **The α-invariant.** Recall that the Hitchin invariant $\alpha(M) \in KO_j(\mathrm{pt})$ of a closed spin j-manifold is the real Dirac index. It depends only on the spin bordism class. We abbreviate $KO_j(\mathrm{pt})$ to KO_j. A homotopy 9-sphere is orientable and has vanishing second Stiefel–Whitney class because $H^2(\Sigma; \mathbb{Z}/2) = 0$. It is therefore spin after an orientation has been fixed, and the corresponding spin structure is unique because $H^1(\Sigma; \mathbb{Z}/2) = 0$. The real Dirac index defines a homomorphism

$$\alpha : \Theta_9 \longrightarrow KO_9 \cong \mathbb{Z}/2.$$

To compute this index from a stable homotopy class, let Ω^{fr}_* and Ω^{Spin}_* denote the bordism rings of closed stably framed and spin manifolds, respectively. Addition is

disjoint union and multiplication is Cartesian product; in each case bordisms carry the same structure. A stable normal framing induces a spin structure on the tangent bundle. The Pontryagin–Thom isomorphism and the map that forgets the framing therefore define the graded-ring homomorphism

$$\alpha^{\mathrm{fr}} : \pi_*^s \cong \Omega_*^{\mathrm{fr}} \longrightarrow \Omega_*^{\mathrm{Spin}} \xrightarrow{\alpha} KO_* \tag{2.2}$$

[4, Section 2.3, equation (2.11)]. The subscript on α_n^{fr} will denote its degree-n part.

In degrees eight and nine this map factors through the stable J-cokernel. Indeed, π_8^s is finite and $KO_8 \cong \mathbb{Z}$, so $\alpha_8^{\mathrm{fr}} = 0$. The generator of $\operatorname{Im} J_9$ in (2.1) is $\eta(\bar{\nu} + \epsilon)$; multiplicativity then gives $\alpha_9^{\mathrm{fr}}(\operatorname{Im} J_9) = 0$. Thus there are induced maps

$$\bar{\alpha}_n : \operatorname{Coker} J_n \longrightarrow KO_n, \qquad \bar{\alpha}_n([x]) = \alpha_n^{\mathrm{fr}}(x), \qquad \alpha(\Sigma) = \bar{\alpha}_n\Big(p_n(\Sigma)\Big), \quad n = 8, 9. \tag{2.3}$$

For the last identity, represent $p_n(\Sigma)$ by the Pontryagin–Thom class of a stable normal framing of Σ, as in the preceding subsection. Forgetting that framing gives the unique spin structure compatible with its orientation. Changing the framing changes the stable class by an element of $\operatorname{Im} J_n$, whose index has just been shown to vanish.

In particular, $\alpha_9^{\mathrm{fr}}(\eta\epsilon) = \alpha_1^{\mathrm{fr}}(\eta)\alpha_8^{\mathrm{fr}}(\epsilon) = 0$. The Adams element μ satisfies $\alpha_9^{\mathrm{fr}}(\mu) = 1 \in KO_9 \cong \mathbb{Z}/2$ [4, Section 2.3]. Also $\alpha(\kappa) = 0$, since κ bounds a parallelizable manifold. Applying (2.3) to the basis (1.5) gives

$$\ker \alpha = \langle \kappa, \lambda \rangle = \{0, \kappa, \lambda, \kappa + \lambda\}. \tag{2.4}$$

By Hitchin's obstruction, the four classes in $\widehat{\mu} + \ker \alpha$ admit no metric of positive scalar curvature. The following proposition gives the corresponding obstruction for nonnegative sectional curvature.

Proposition 2.1 (The nonnegative-sectional-curvature obstruction)**.** *If a smooth homotopy 9-sphere Σ has $\alpha(\Sigma) \neq 0$, then it admits no Riemannian metric of nonnegative sectional curvature.*

Proof. This implication is also recorded in [3, Proposition 7.1]. Suppose $\sec_g \geq 0$. Let D_g be the spin Dirac operator. An invertible Dirac operator has zero real index, so $\alpha(\Sigma) \neq 0$ implies that $\ker D_g$ contains a nonzero harmonic spinor ψ. The Lichnerowicz formula [10] gives

$$0 = \int_\Sigma \left(|\nabla \psi|^2 + \frac{\mathrm{Scal}_g}{4} |\psi|^2 \right) dV_g.$$

Both terms are nonnegative. Hence ψ is parallel with constant nonzero norm, and Scal_g vanishes identically. Complete an orthonormal basis of any prescribed two-plane to a tangent orthonormal basis. The identity $\mathrm{Scal}_g = 2\sum_{i<j} \sec_g(e_i \wedge e_j)$ then forces that plane's sectional curvature to vanish. Thus g is flat. Completeness and simple connectivity would make (Σ, g) isometric to $\mathbb{R}^9$, contradicting compactness. □

Accordingly, the proof of theorem 1.1 reduces to constructing positive-sectional-curvature metrics on κ, λ, and $\kappa + \lambda$.

2.3. **Gluing discs with fixed parametrizations.** For each smooth n-disc D_i, fix a diffeomorphism $\chi_i : D^n \to D_i$ and give D_i the transported standard orientation. The restriction $\partial\chi_i : S^{n-1} \to \partial D_i$ parametrizes its boundary.

Given an orientation-preserving diffeomorphism $\varphi : S^{n-1} \to S^{n-1}$, glue the two discs by identifying $\chi_N(y)$ with $\chi_S(\varphi(y))$. The southern disc has the opposite orientation, as in the clutching convention above. For $n = 9$ we write

$$\Sigma = D^9_N \cup_\varphi D^9_S, \tag{2.5}$$

with this orientation convention understood. We also use φ for the actual boundary map $\partial\chi_S \circ \varphi \circ (\partial\chi_N)^{-1}$ when the domain and target are the boundaries of the parametrized discs.

For metrics g_i on D_i, let h_i and B_i be their boundary metrics and outward second fundamental forms, expressed on the common S^{n-1} by $\partial\chi_i$. Thus the boundary isometry condition is exactly $h_N = \varphi^* h_S$. Replacing χ_i by $\chi_i \circ u_i$, where $u_i : D^n \to D^n$ is an orientation-preserving diffeomorphism, replaces the coordinate attaching map by

$$\varphi' = (\partial u_S)^{-1} \circ \varphi \circ \partial u_N.$$

Put $v_i = \partial u_i$. The boundary tensors in the new coordinates are $h'_i = v_i^* h_i$ and $B'_i = v_i^* B_i$. Since $v_S \circ \varphi' = \varphi \circ v_N$, functoriality of pullback gives

$$h'_N - \varphi'^* h'_S = v_N^*(h_N - \varphi^* h_S), \qquad B'_N + \varphi'^* B'_S = v_N^*(B_N + \varphi^* B_S).$$

Thus changing the disc parametrizations preserves both gluing conditions.

2.4. **Group actions and relative diffeomorphisms.** For a Euclidean space W, put $S(W) = \{x \in W : |x| = 1\}$ and $D(W) = \{x \in W : |x| \le 1\}$. For $r > 0$, write $D^j(r) = \{x \in \mathbb{R}^j : |x| \le r\}$ and $D^j = D^j(1)$. The symbols ∂ and Int denote boundary and interior. On Euclidean spaces, $\langle \cdot, \cdot \rangle$ denotes the specified Euclidean inner product. For a linear map K between inner-product spaces, adjoints and operator norms use those inner products; in particular K^* satisfies $\langle K\xi, Y\rangle = \langle \xi, K^* Y\rangle$. For vectors X, Y we use the exterior-product norm $|X \wedge Y|^2 = |X|^2|Y|^2 - \langle X, Y\rangle^2$. We identify S^3 with the unit quaternions and its Lie algebra with $\operatorname{Im}\mathbb{H}$, with $[\xi, \zeta] = \xi\zeta - \zeta\xi$. For $q \in S^3$, write $L_q(z) = qz$, $R_q(z) = zq$ and $\operatorname{Ad}_q \xi = q\xi q^{-1}$. The ordered basis (i, j, k) fixes the Euclidean metric and orientation of $\operatorname{Im}\mathbb{H}$. All metrics denoted by $h_{S(W)}$ are unit round metrics. If $\rho : S^3 \to O(W)$ is an orthogonal representation, a map $\beta : S(W) \to S^3$ is *invariant* if $\beta(\rho(q)y) = \beta(y)$, and *conjugation equivariant* if $\beta(\rho(q)y) = q\beta(y)q^{-1}$. The associated map $F_\beta : S(W) \to S(W)$ is defined by $F_\beta(y) = \rho(\beta(y))y$. We write J_β in the equivariant case and L_β in the invariant case.

For a compact smooth manifold M with boundary, let $\operatorname{Diff}_\partial(M) = \operatorname{Diff}(M, \operatorname{rel}\partial)$ be the group of smooth diffeomorphisms equal to the identity near ∂M, with composition and the C^∞ topology. A collar is a smooth embedding $c : \partial M \times [0, \varepsilon) \to M$ with $c(x, 0) = x$. Each element of $\operatorname{Diff}_\partial(M)$ fixes a collar pointwise. For each compact parameter family we choose a common collar, fixed pointwise for diffeomorphisms and mapped to 1 for relative maps into S^3. Relative collar compression and smoothing give such representatives for the homotopy classes used below. A map described as smooth on a manifold with boundary is understood to be smooth up to the boundary. Two relative diffeomorphisms f_0, f_1 of M are pseudo-isotopic if there is a diffeomorphism $F : M \times [0, 1] \to M \times [0, 1]$ such that $F(x, i) = (f_i(x), i)$ for $i = 0, 1$ and F is the identity near $\partial M \times [0, 1]$. Unlike an isotopy, F need not preserve the intermediate slices. We denote the group of these classes by $\widetilde{\pi}_0 \operatorname{Diff}(M, \operatorname{rel}\partial)$. The basepoint of a diffeomorphism group is its identity. For based spaces X, Y, $\operatorname{Map}_*(X, Y)$ denotes the space of continuous basepoint-preserving maps, with the compact-open topology. We write $\Omega^j Y = \operatorname{Map}_*(I^j/\partial I^j, Y)$, where $I = [0, 1]$; adjunction gives $\pi_i(\Omega^j Y) = \pi_{i+j}(Y)$

at the constant map. Other components and base maps will be specified explicitly. The smash product is $X \wedge Y = (X \times Y)/(X \times \{*\} \cup \{*\} \times Y)$. For a topological group G, the notation $EG \to BG$ denotes a universal principal bundle with contractible total space EG.

Lemma 2.2 (Diffeomorphisms associated with group actions)**.** *Let V be an orthogonal S^3-representation with action ρ. Suppose $\beta : S(V) \to S^3$ is smooth and either invariant or conjugation equivariant. Then*

$$F_\beta(y) = \rho(\beta(y))y$$

is a smooth diffeomorphism with inverse $F_{\beta^{-1}}$, where $\beta^{-1}(y) = \beta(y)^{-1}$. Its signed Jacobian for the round volume form is identically one. A smooth homotopy through invariant or conjugation-equivariant maps gives an isotopy of the corresponding diffeomorphisms.

Proof. In either case $\beta(F_\beta(y)) = \beta(y)$; in the conjugation-equivariant case this is the identity $\beta(y)\beta(y)\beta(y)^{-1} = \beta(y)$. This proves the inverse formula. Set $K_y\xi = \frac{d}{ds}\Big|_{s=0} \rho(e^{s\xi})y$ and $A_y = \beta(y)^{-1}d\beta_y$. Differentiation gives

$$dF_\beta|_y = \rho(\beta(y))\Big(I + K_y A_y\Big).$$

The orthogonal factor preserves the oriented round volume, since S^3 is connected. By the rectangular determinant identity,

$$\det(I + K_y A_y) = \det(I + A_y K_y).$$

If β is invariant, $A_y K_y = 0$. If β is conjugation equivariant,

$$A_y K_y = \mathrm{Ad}_{\beta(y)^{-1}} - I.$$

The determinant is 1 in either case, and the inverse formula is smooth in a homotopy parameter. □

2.5. **Bundle conventions.** A principal right G-bundle $\pi : P \to B$ has a free right action $R_h(p) = ph$ that is transitive on each fiber. Its local trivializations are equivariant, with action $(x, v)h = (x, vh)$. The pullback of a bundle $p : E \to Y$ by a map $f : X \to Y$ is $f^*E = \{(x, e) \in X \times E : f(x) = p(e)\}$, with projection to X.

For $G = S^3$, a principal connection is an $\operatorname{Im}\mathbb{H}$-valued one-form ω on P satisfying $\omega(\xi^\#) = \xi$ and $R_h^*\omega = \mathrm{Ad}_{h^{-1}}\omega$, where $\xi_p^\# = \frac{d}{dt}\Big|_{t=0} p\exp(t\xi)$. Its horizontal space is $\ker\omega$, and its vertical space is $\ker d\pi$. For a local section s_i write $A_i = s_i^*\omega$; in coordinates $p = s_i(x)v_i$,

$$\omega_i = \mathrm{Ad}_{v_i^{-1}} A_i + v_i^{-1}dv_i.$$

The curvature is $d\omega + \frac{1}{2}[\omega \wedge \omega]$, where $[\omega \wedge \omega](U, V) = 2[\omega(U), \omega(V)]$. It is an equivariant horizontal form and hence corresponds to a 2-form on B with values in the adjoint bundle

$$\operatorname{ad} P = (P \times \operatorname{Im}\mathbb{H})/\Big((p, \xi) \sim (ph, \mathrm{Ad}_{h^{-1}}\xi)\Big).$$

An auxiliary left group action on a total space will be denoted by $\star$; its quotient $P/G_\star$ is the space of its orbits $[p]_\star = \{g \star p : g \in G\}$.

3. Diffeomorphisms supported in a tubular neighborhood

3.1. **Left and right quaternionic multiplication.** Fix $0 < c_0 < c_1 < 1$ and give $\mathbb{R}^5 = \mathbb{R} \oplus \mathbb{H}$ the product orientation. Choose a smooth map

$$b : \mathbb{R}^5 \longrightarrow S^3, \qquad b(u) = 1 \quad \text{for } |u| \geq c_0, \tag{3.1}$$

whose restriction to $D^5(c_1)$ descends to a based map on $D^5(c_1)/\partial D^5(c_1) \cong S^5$ representing

$$[b] = \eta_3\eta_4 \in \pi_5(S^3) \cong \mathbb{Z}/2. \tag{3.2}$$

Here $\eta_2 : S^3 \to S^2 \subset \operatorname{Im}\mathbb{H}$ is the Hopf map $\eta_2(q) = qiq^{-1}$, and $\eta_j = \Sigma^{j-2}\eta_2$ for $j \geq 2$; products of maps are composed from right to left. The nonzero class in (3.2) is the classical calculation in [20]. A representative can be made constant on a boundary collar by a smooth radial collapse and relative smoothing, and then extended by 1 to $\mathbb{R}^5$. The oriented radial identification with the unit disc fixes the identification of its relative class with $\pi_5(S^3)$.

Put $T = S^3 \times D^5(c_1)$ and fix this b. Smooth homotopies of b that are equal to 1 on a common boundary neighborhood induce isotopies of the diffeomorphisms defined below. Collar straightening identifies these relative pseudo-isotopy classes with those used by Krylov [12]. Define relative diffeomorphisms

$$R(q,u) = (qb(u), u), \qquad L(q,u) = (b(u)q, u). \tag{3.3}$$

They are the identity near ∂T and satisfy $RL = LR$.

Let

$$\mathcal{E} = \widetilde{\pi}_0 \operatorname{Diff}(T, \operatorname{rel}\partial) \tag{3.4}$$

be the group of relative pseudo-isotopy classes. Composition in $\mathcal{E}$ is written multiplicatively; we also write R, L for the classes of the two diffeomorphisms in (3.3).

Embed T as a tubular neighborhood in S^8 by

$$e_0(q,u) = \left(\sqrt{1-|u|^2}\, q, u\right) \in S(\mathbb{H} \oplus \mathbb{R}^5), \tag{3.5}$$

where $|u| \leq c_1 < 1$. Its inverse recovers $q = w/|w|$ and u; the quaternionic coordinate is nonzero on $e_0(T)$. For $f \in \operatorname{Diff}_\partial(T)$ define

$$\widehat{f}(x) = \begin{cases} e_0 \circ f \circ e_0^{-1}(x), & x \in e_0(T), \\ x, & x \notin e_0(T). \end{cases}$$

The two formulas agree on a neighborhood of $\partial e_0(T)$. Thus $\widehat{f}$ is a smooth orientation-preserving diffeomorphism. Gluing two discs along $\widehat{f}$ defines the homomorphism

$$\Phi : \mathcal{E} \longrightarrow \Theta_9, \qquad \Phi([f]) = [\Sigma_{\widehat{f}}]. \tag{3.6}$$

To check that Φ is well-defined, extend a relative pseudo-isotopy to $H : S^8 \times I \to S^8 \times I$, with endpoint maps $\widehat{f}_0, \widehat{f}_1$. The diffeomorphism $(\widehat{f}_1^{-1} \times \mathrm{id}_I) \circ H$ restricts to $\widehat{f}_1^{-1}\widehat{f}_0$ at the bottom and to the identity at the top. Straighten it to the identity on a top collar and cap that end by a 9-disc, extending by the identity. The capped cylinder is a 9-disc, so $\widehat{f}_1^{-1}\widehat{f}_0$ extends over it. The clutching convention of Section 2 then gives $[\Sigma_{\widehat{f}_0}] = [\Sigma_{\widehat{f}_1}]$. Extension preserves composition, which represents connected sum under this convention; hence Φ is a homomorphism.

Choose an orientation-preserving embedding $e : D^8 \hookrightarrow \operatorname{Int} T$. The disc-supported clutching identification $\widetilde{\pi}_0 \operatorname{Diff}_\partial(D^8) \cong \Theta_9$ lets us represent $w \in \Theta_9$ by $f_w \in \operatorname{Diff}_\partial(D^8)$

[11, 1]. Extend ef_we^{-1} by the identity outside $e(D^8)$ and denote its class in $\mathcal{E}$ by $\iota(w)$. Relative pseudo-isotopies extend in the same way, so this defines a homomorphism

$$\iota : \Theta_9 \longrightarrow \mathcal{E}. \tag{3.7}$$

It is injective because, with consistent orientation conventions,

$$\Phi \circ \iota = \mathrm{id}_{\Theta_9}. \tag{3.8}$$

Indeed, extension first to T and then to S^8 gives the disc-supported clutching representative of the same oriented sphere.

Lemma 3.1 (Conjugation on diffeomorphisms supported in a disc)**.** *The subgroup $\mathcal{N} = \iota(\Theta_9)$ is central in $\mathcal{E}$. More generally, conjugation by any orientation-preserving diffeomorphism of T fixes $\mathcal{N}$ pointwise, even when that diffeomorphism is not fixed on ∂T.*

Proof. Represent an element of $\mathcal{N}$ by a disc diffeomorphism supported in the image of an orientation-preserving embedding $e : D^8 \hookrightarrow \operatorname{Int} T$. Conjugation by g replaces e by $g \circ e$ and leaves the disc representative itself unchanged.

Shrink both parametrized embeddings to sufficiently small concentric discs. In coordinate neighborhoods each can be deformed to its orientation-preserving derivative frame. The oriented frame bundle of the connected manifold $\operatorname{Int} T$ is connected, so a path of small framed discs connects the two embeddings. The whole path has compact image in $\operatorname{Int} T$. Isotopy extension gives an ambient isotopy fixed near ∂T. Conjugating the supported diffeomorphism along that isotopy preserves its relative mapping class. Thus conjugation fixes $\mathcal{N}$ pointwise. Taking g to be any relative diffeomorphism proves centrality. □

Lemma 3.2. *The diffeomorphisms $\widehat{R}, \widehat{L} : S^8 \to S^8$ both extend over D^9. In particular,*

$$\Phi(R) = \Phi(L) = 0.$$

Proof. Regard

$$D^9 = D(\mathbb{H} \oplus \mathbb{R}^5) = \{(w, u) : |w|^2 + |u|^2 \leq 1\}.$$

The maps

$$(w, u) \longmapsto (wb(u), u), \qquad (w, u) \longmapsto (b(u)w, u)$$

are smooth norm-preserving diffeomorphisms with explicit inverses obtained by replacing b with b^{-1}. Their restrictions to S^8 are the identity extensions of R and L under (3.5). □

3.2. **The relative mapping-class quotient.** For $q > 2$, let C_n^q denote the group of smooth isotopy classes of parametrized embeddings $S^n \hookrightarrow S^{n+q}$, with connected sum as operation. A normal framing of an embedding e is an oriented trivialization of its normal bundle e^*TS^{n+q}/TS^n. The group FC_n^q consists of embeddings equipped with such a framing, up to isotopy through framed embeddings; its operation is framed connected sum. The superscript records codimension. Equivalently, a framed embedding is given by an oriented tubular embedding $S^n \times D^q \hookrightarrow S^{n+q}$. Concordance and isotopy give the same classes in this range [8, Theorem 1.2 and Section 5]. Since $\mathcal{N}$ is central by lemma 3.1, we may form the quotient $\mathcal{Q} = \mathcal{E}/\mathcal{N}$. Write $\overline{f} = f\mathcal{N}$ for the image of $f \in \mathcal{E}$ in $\mathcal{Q}$.

Theorem 3.3 (Relative mapping classes and a cardinality bound)**.** *The quotient by* $\mathcal{N} = \iota(\Theta_9)$ *satisfies*

$$\mathcal{Q} := \mathcal{E}/\mathcal{N} \cong FC_5^4, \qquad |\mathcal{Q}| \leq 8. \tag{3.9}$$

Proof. For $(p, q) = (3, 5)$, Krylov's relative classification [12, Lemma 1 and its proof] gives $\mathcal{E} \cong \Theta_9 \oplus FC_5^4$. Its Θ_9 summand is the image of ι, represented by diffeomorphisms supported in a disc. Quotienting by this summand gives $\mathcal{Q} \cong FC_5^4$.

Haefliger's calculation gives $C_5^4 \cong \mathbb{Z}/2$ [8, Corollary 8.14]. Fix a representative of either unframed class. If its oriented normal bundle is not trivial, it has no framing. Otherwise, relative to one framing, the homotopy classes of oriented normal framings form a torsor for

$$[S^5, SO(4)] = \pi_5 SO(4) \cong \pi_5(S^3 \times S^3) \cong (\mathbb{Z}/2)^2.$$

Here the covering $S^3 \times S^3 \to SO(4)$ induces the isomorphism. An isotopy of the underlying embedding transports its normal framing to the chosen representative. Isotopy extension and uniqueness of tubular neighborhoods identify these data with tubular embeddings $S^5 \times D^4 \hookrightarrow S^9$. Self-isotopies may identify different framings. There are therefore at most four framed classes over each of the two unframed classes, proving the bound. □

3.3. **Changing the normal framing.** Identify $\mathbb{R}^5 = \mathbb{R} \oplus \mathbb{H}$ and define

$$\tau_1 : S^3 \longrightarrow SO(5), \qquad \tau_1(q) = I_{\mathbb{R}} \oplus L_q. \tag{3.10}$$

The based map τ_1 changes the normal coordinates through the diffeomorphism

$$T_\tau(q, u) = (q, \tau_1(q)u). \tag{3.11}$$

Conjugation by T_τ preserves $\mathrm{Diff}(T, \mathrm{rel}\,\partial)$ and induces

$$\mathcal{A} : \mathcal{E} \longrightarrow \mathcal{E}, \qquad \mathcal{A}[f] = [T_\tau f T_\tau^{-1}]. \tag{3.12}$$

The map T_τ preserves orientation: its derivative has diagonal blocks I and $\tau_1(q) \in SO(5)$. Lemma 3.1 therefore gives $\mathcal{A} \circ \iota = \iota$. It induces an automorphism $\overline{\mathcal{A}}$ of $\mathcal{Q}$, defined by $\overline{\mathcal{A}}(\overline{f}) = \overline{\mathcal{A}(f)}$.

A second framing uses $\mathbb{R}^5 = \mathbb{R}^2 \oplus \mathrm{Im}\,\mathbb{H}$:

$$\tau_2(q) = I_{\mathbb{R}^2} \oplus \mathrm{Ad}_q \in SO(5). \tag{3.13}$$

Lemma 3.4. *In* $\pi_3 SO(5) \cong \mathbb{Z}$*,*

$$[\tau_2] = 2[\tau_1]. \tag{3.14}$$

Consequently conjugation by $(q, u) \mapsto (q, \tau_2(q)u)$ *induces* $\mathcal{A}^2$ *on* $\mathcal{E}$*.*

Proof. Identify $\mathbb{R}^5 = \mathbb{R} \oplus \mathbb{H} = \mathbb{R} \oplus \mathbb{R} \oplus \mathrm{Im}\,\mathbb{H}$ in the stated coordinate order. Let $c(z) = \overline{z}$ and put $J_0 = (-I_{\mathbb{R}}) \oplus c \in SO(5)$. Both factors reverse orientation, so their direct sum preserves it. Right multiplication by $\overline{q}$ is $cL_q c$, and hence

$$J_0 \tau_1(q) J_0^{-1} = I_{\mathbb{R}} \oplus R_{\overline{q}}, \qquad \tau_1(q) J_0 \tau_1(q) J_0^{-1} = \tau_2(q).$$

Here $R_{\overline{q}}(z) = z\overline{q}$. Choose $U_t \in SO(5)$, $0 \leq t \leq 1$, by rotation through angle πt in each of the ordered orthogonal planes $\mathrm{span}\{(1, 0), (0, i)\}$ and $\mathrm{span}\{(0, j), (0, k)\}$, fixing $(0, 1)$. Then $U_0 = I$ and $U_1 = J_0$. The formula

$$H_t(q) = \tau_1(q) U_t \tau_1(q) U_t^{-1}$$

defines a smooth based homotopy in $SO(5)$, with

$$H_0(q) = \tau_1(q)^2, \qquad H_1(q) = \tau_2(q), \qquad H_t(1) = I.$$

Pointwise multiplication represents addition in $\pi_3 SO(5)$, proving (3.14) with the indicated sign.

The maps $T_t(q,u) = (q, H_t(q)u)$ are diffeomorphisms of T preserving $|u|$ and fixing $S^3 \times \{0\}$. For every relative diffeomorphism f, the family $T_t f T_t^{-1}$ is an isotopy relative to a common boundary collar. At $t = 0$ it is conjugation by T_τ^2; at $t = 1$ it is conjugation by $(q,u) \mapsto (q, \tau_2(q)u)$. Their actions on $\mathcal{E}$ therefore agree. □

3.4. **Homotopy classes relative to the boundary.** For this relative-homotopy computation, identify the unit D^5 with $D^5(c_1)$ by $u \mapsto c_1 u$ and transport the maps by this identification. We retain the symbols T, b, R, L in these rescaled coordinates. Let $\mathcal{H}$ be the homotopy classes of continuous self-maps of $T = S^3 \times D^5$ that restrict to the identity on ∂T, where every map in the homotopy has this same boundary restriction. Composition initially makes $\mathcal{H}$ a monoid.

Lemma 3.5 (The relative-homotopy quotient)**.** *The four classes of $\mathcal{H}$ are represented by* $\mathrm{id}, R, L, LR^{-1}$*, and their composition group is $(\mathbb{Z}/2)^2$. Forgetting smoothness gives a surjective homomorphism*

$$\mathcal{Q} \longrightarrow \mathcal{H}, \qquad \mathcal{K} := \ker(\mathcal{Q} \to \mathcal{H}), \qquad |\mathcal{K}| \le 2. \tag{3.15}$$

Proof. Write a relative map as $f = (F, G)$. Convex interpolation of G with u stays in D^5 and is fixed on the boundary. Define

$$e_f(u) = F(1,u), \qquad a_f(q,u) = e_f(u)^{-1} q^{-1} F(q,u). \tag{3.16}$$

The first component is a based map on $D^5/\partial D^5 = S^5$. The second is constant on $S^3 \times \partial D^5$ and $\{1\} \times D^5$, hence gives a based map on $S^3 \wedge S^5 = S^8$. Conversely $F(q,u) = q e_f(u) a_f(q,u)$ reconstructs F. These constructions apply to homotopies as well, giving a set bijection

$$\mathcal{H} \longleftrightarrow \pi_5(S^3) \times \pi_8(S^3) \cong (\mathbb{Z}/2)^2.$$

Put $\beta = [b]$ and $\chi = \langle \beta, \iota_3 \rangle \in \pi_8(S^3)$, where ι_3 is the class of the identity map of S^3. For based maps $a : S^p \to G$ and $d : S^q \to G$ into a topological group, the Samelson product $\langle [a], [d] \rangle \in \pi_{p+q}(G)$ is the class induced on $S^p \wedge S^q$ by $(x,y) \mapsto a(x) d(y) a(x)^{-1} d(y)^{-1}$. The commutator is the identity on the wedge, so this is a based map. Both β and χ are nonzero. For the second assertion, naturality gives

$$\chi = \pm \langle \iota_3, \iota_3 \rangle \circ \eta_6 \eta_7. \tag{3.17}$$

The Samelson square generates $\pi_6(S^3) \cong \mathbb{Z}/12$ [22, proof of Proposition III.1]; its 2-primary part is an odd multiple of Toda's order-four class $\nu' \in \pi_6(S^3)$. The nonzero composite

$$\nu' \eta_6 \eta_7 \neq 0 \tag{3.18}$$

is [20, Proposition 5.9]. The pointwise coordinates of the maps L and LR^{-1} are

$$e_L(u) = b(u), \qquad a_L(q,u) = b(u)^{-1} q^{-1} b(u) q,$$
$$e_{LR^{-1}}(u) = 1, \qquad a_{LR^{-1}}(q,u) = q^{-1} b(u) q b(u)^{-1}.$$

The two displayed second coordinates are commutators of based maps. Inversion represents negation in their homotopy groups, and interchanging the smash factors changes only the sign of the resulting class. Bilinearity of the Samelson product

therefore identifies both second coordinates with χ, since $\pi_8(S^3)$ has order two. Together with $e_R = b$ and $a_R = 1$, this gives

f	id	R	L	LR^{-1}
$([e_f], [a_f])$	$(0,0)$	$(\beta, 0)$	(β, χ)	$(0, \chi)$.

The maps R, L commute exactly, and pointwise multiplication represents addition in $\pi_5(S^3)$. Since this group has order two, b^2 admits a null homotopy through maps equal to 1 on a common boundary neighborhood. Thus R, L are involutions in $\mathcal{E}$, and their four products exhaust $\mathcal{H}$ and identify its composition law with $(\mathbb{Z}/2)^2$.

A relative pseudo-isotopy induces a relative homotopy by projecting its target $T \times I$ onto T. If $f_w \in \mathrm{Diff}_\partial(D^8)$ represents $w \in \Theta_9$, the maps

$$h_t(x) = (1-t)f_w(x) + tx, \qquad 0 \le t \le 1,$$

stay in the convex disc and are the identity on its boundary collar. Transport them to the supporting disc in T and extend by the identity. This relative continuous homotopy shows that $\mathcal{E} \to \mathcal{H}$ kills $\mathcal{N}$ and factors through $\mathcal{Q}$. Its surjectivity and $|\mathcal{Q}| \le 8$ imply (3.15). □

Lemma 3.6 (The action on relative homotopy classes)**.** *The automorphism $\mathcal{A}$ induces the identity on $\mathcal{H}$.*

Proof. Let $\bar{b} : S^5 = D^5/\partial D^5 \to S^3$ be the based map induced by b. Its class is $\eta_3\eta_4$. Write E for suspension and let ν_5 generate the 2-primary part of $\pi_8(S^5)$. Toda's relations

$$\eta_3 E\nu' = 0, \qquad \eta_4\nu_5 = (E\nu')\eta_7$$

from [20, Lemma 5.7 and equations (5.9)] give $\bar{b}\nu_5 = \eta_3\eta_4\nu_5 = 0$. The odd-primary subgroup of $\pi_8(S^5)$ also maps to zero, since $\pi_8(S^3) = \mathbb{Z}/2$. Hence

$$\bar{b}_* : \pi_8(S^5) \longrightarrow \pi_8(S^3) \quad \text{is zero.} \tag{3.19}$$

The rotations of D^5 descend to a based family $q \mapsto r_q$ in $\mathrm{Map}_*(S^5, S^5)$, where $r_q([u]) = [\tau_1(q)^{-1}u]$ and $r_1 = \mathrm{id}$. View these mapping spaces as fivefold loop spaces, with base maps id and $\bar{b}$, respectively. Translation by a loop is a homotopy equivalence between components, and postcomposition preserves loop concatenation. If $*$ denotes concatenation, then

$$\Omega^5\bar{b}(\mathrm{id} * \zeta) = \bar{b} * \left(\Omega^5\bar{b}(\zeta)\right).$$

Together with the standard unit homotopies, this identity identifies the induced map on π_3 at $(\mathrm{id}, \bar{b})$ with the homomorphism in (3.19). Thus $q \mapsto \bar{b} \circ r_q$ has a based null homotopy with base map $\bar{b}$.

Set $R_\tau = T_\tau R T_\tau^{-1}$ and $L_\tau = T_\tau L T_\tau^{-1}$, which represent $\mathcal{A}R$ and $\mathcal{A}L$. Their first components are

$$\mathrm{pr}_1 R_\tau(q,u) = qb(\tau_1(q)^{-1}u), \qquad \mathrm{pr}_1 L_\tau(q,u) = b(\tau_1(q)^{-1}u)q.$$

Pulling the preceding based null homotopy back to D^5 changes these components to $qb(u)$ and $b(u)q$ while fixing ∂T. Convex interpolation changes each second component to u, also relative to ∂T. Hence $R_\tau \simeq_\partial R$ and $L_\tau \simeq_\partial L$ as continuous maps. Since the classes of R, L generate $\mathcal{H}$, the induced action on $\mathcal{H}$ is the identity. □

Lemma 3.7 (The induced action on $\mathcal{Q}$)**.** *The induced automorphism $\overline{\mathcal{A}}$ on $\mathcal{Q}$ satisfies $\overline{\mathcal{A}}^2 = \mathrm{id}$.*

Proof. The kernel $\mathcal{K}$ has order one or two and is invariant under $\overline{\mathcal{A}}$, so it is fixed pointwise. Since the action on $\mathcal{H}$ is trivial, write $\overline{\mathcal{A}}(x) = xk_x$ for $x \in \mathcal{Q}$, with $k_x \in \mathcal{K}$. Then

$$\overline{\mathcal{A}}^2(x) = \overline{\mathcal{A}}(x)\overline{\mathcal{A}}(k_x) = xk_x^2 = x.$$

□

4. The classes κ and λ

We identify the sphere obtained using τ_1 with the Kervaire sphere. The framing τ_2 gives a class λ whose normal invariant is $[\eta\epsilon]$. Both constructions use the map b from (3.1).

4.1. **The Kervaire class.** Consider the orthogonal representation

$$V_\kappa = \mathbb{R} \oplus \mathbb{H} \oplus \mathbb{H}, \qquad \rho_\kappa(q)(t, w, z) = (t, qw, qz). \tag{4.1}$$

On the open set $w \neq 0$, where the action is free, write

$$q = \frac{w}{|w|}, \qquad u = (t, q^{-1}z) \in \operatorname{Int} D^5(1).$$

Restricting these coordinates to $|u| \leq c_1$ gives the tubular embedding

$$e_\kappa(q, t, z) = \left(t, \sqrt{1 - t^2 - |z|^2}\, q, qz\right). \tag{4.2}$$

Let $\Pi_\kappa : \mathbb{H} \oplus (\mathbb{R} \oplus \mathbb{H}) \to V_\kappa$ be $\Pi_\kappa(w, (t, v)) = (t, w, v)$. This orthogonal permutation preserves orientation, since it moves a real line past a four-dimensional block. With the same symbol for its restriction to the unit sphere,

$$e_\kappa = \Pi_\kappa \circ e_0 \circ T_\tau. \tag{4.3}$$

The map Π_κ extends linearly over the 9-disc. Conjugating an attaching map by this coordinate change therefore preserves its oriented homotopy-sphere class.

Define a conjugation-equivariant map on $e_\kappa(T)$ by

$$\beta_\kappa(e_\kappa(q, u)) = qb(u)q^{-1}, \tag{4.4}$$

and extend it by 1 outside $e_\kappa(T)$. The associated clutching diffeomorphism is

$$J_{\beta_\kappa}(y) = \rho_\kappa(\beta_\kappa(y))y. \tag{4.5}$$

For $y = e_\kappa(q, u)$, equivariance of e_κ gives $\rho_\kappa(h)e_\kappa(q, u) = e_\kappa(hq, u)$. Taking $h = qb(u)q^{-1}$ therefore yields

$$e_\kappa^{-1} \circ J_{\beta_\kappa} \circ e_\kappa(q, u) = (qb(u), u) = R(q, u).$$

Together with (4.3), this gives

$$[\Sigma_{J_{\beta_\kappa}}] = \Phi(\mathcal{A}R). \tag{4.6}$$

A conjugation-equivariant map on $S(V_\kappa)$ determines a path in the gauge group of the quaternionic Hopf bundle. We first determine the endpoint values and then the fundamental group of this gauge group.

On $S^7 = S(\mathbb{H} \oplus \mathbb{H})$ use the free left action $q(w, z) = (qw, qz)$. Its quotient is $S^4 = S(\mathbb{R} \oplus \mathbb{H})$, with Hopf projection

$$p_H(w, z) = (|w|^2 - |z|^2,\, 2\overline{w}z).$$

We use the equivariant-function model of its gauge group,

$$\mathcal{G}_H = \{g \in C^\infty(S^7, S^3) : g(qx) = qg(x)q^{-1}\},$$

with pointwise multiplication, identity $g \equiv 1$, and the C^∞ topology. The corresponding bundle automorphism is $x \mapsto g(x)^{-1}x$; this convention identifies pointwise multiplication with composition of these automorphisms. After the radial identification of each nonpolar latitude with S^7, a smooth conjugation-equivariant map on $S(\mathbb{R} \oplus \mathbb{H}^2)$ determines a path of gauge transformations. At either pole its value lies in the center $\{1, -1\}$.

Lemma 4.1. *The constant gauge transformations* 1 *and* -1 *lie in distinct path components of* $\mathcal{G}_H$.

Proof. Trivialize the Hopf bundle over the two hemispheres of S^4, with equatorial transition $c(q) = q$. If a path from 1 to -1 in $\mathcal{G}_H$ existed, it would be represented by maps $a_N, a_S : D^4 \times I \to S^3$, constant at the two ends, with

$$a_N(q, s) = q a_S(q, s) q^{-1} \qquad (q \in S^3).$$

Contract the southern D^4 to an interior point. This deforms a_S, relative to its two end faces, to a path independent of the disc coordinate. Since $\pi_1(S^3) = 0$, that path can then be deformed, still relative to its endpoints, to $a_S(s) = \exp(\pi s i)$. On the common side face the corresponding homotopy of a_N is prescribed by conjugation. The homotopy extension property for the boundary of $D^4 \times I$ extends it over the northern cylinder. The resulting map on the boundary of $D^4 \times I$ has side restriction

$$(q, s) \longmapsto q \exp(\pi s i) q^{-1} = \cos(\pi s) + \sin(\pi s) q i q^{-1}.$$

After collapsing the top and bottom, this is the suspension of the Hopf map $q \mapsto qiq^{-1}$ and represents the nonzero element $\eta_3 \in \pi_4(S^3)$. It therefore cannot extend over $D^4 \times I$. □

It follows that an equivariant transition has the same value at both poles. If this value is -1, multiplication by the central constant -1 changes J_β by the linear map $(t, w, z) \mapsto (t, -w, -z)$, which is orientation preserving and extends over D^9. Thus every such transition may be normalized to a based loop in $\mathcal{G}_H$ without changing its homotopy-sphere class. A preliminary reparameterization makes it constant on small polar caps. Continuous homotopies of these equivariant maps admit relative equivariant smoothing. More precisely, let $f : M \times I \to S^3 \subset \mathbb{H}$ be such a continuous homotopy on a compact S^3-manifold with action $h \cdot x$, stationary near its parameter endpoints and smooth near the invariant closed subset to be fixed. Relative smooth approximation gives $u : M \times I \to \mathbb{H}$, equal to f on a smaller neighborhood of that subset, with $\|u - f\|_\infty < 1/2$. Using normalized Haar measure, set

$$u_{\mathrm{av}}(x, t) = \int_{S^3} h^{-1} u(h \cdot x, t) h \, dh.$$

The average is smooth and equivariant, preserves the prescribed values, and satisfies $\|u_{\mathrm{av}} - f\|_\infty < 1/2$. Thus $u_{\mathrm{av}}/|u_{\mathrm{av}}|$ is the required smooth equivariant homotopy. Applied relative to the polar caps, this shows that the gauge homotopy classes determine smooth isotopy classes of the attaching maps.

Lemma 4.2. *With the identity gauge transformation as basepoint, the gauge group satisfies*

$$\pi_1(\mathcal{G}_H) \cong \pi_5(S^3) \cong \mathbb{Z}/2.$$

In the coordinates defined by e_κ, *the map* b *in* (3.1) *determines the nonzero element of* $\pi_1(\mathcal{G}_H)$.

Proof. Fix $x_* \in S^7$ over a point in the complementary Hopf chart and set

$$\mathrm{ev}_{x_*} : \mathcal{G}_H \to S^3, \qquad g \mapsto g(x_*), \qquad \mathcal{G}_{H,*} = \mathrm{ev}_{x_*}^{-1}(1).$$

The evaluation map is a fibration, and its fiber has weak homotopy type $\mathcal{G}_{H,*} \simeq \Omega^4 S^3$ [22, Lemmas II.7–II.8]. Relative smoothing identifies the homotopy groups of the smooth and continuous models. The long exact sequence gives

$$0 = \pi_2(S^3) \longrightarrow \pi_5(S^3) \longrightarrow \pi_1(\mathcal{G}_H) \longrightarrow \pi_1(S^3) = 0.$$

A gauge transformation supported in the other, trivializing four-disc lies in $\mathcal{G}_{H,*}$ and is represented there by a map $(D^4, \partial D^4) \to (S^3, 1)$. A loop of these transformations is a map $(I \times D^4, \partial(I \times D^4)) \to (S^3, 1)$ and realizes the left-hand homomorphism from $\pi_5(S^3)$ in the displayed exact sequence. The following coordinate change identifies this loop with b in (4.4). On a latitude of $S(\mathbb{R} \oplus \mathbb{H}^2)$ the Hopf chart $w \neq 0$ has affine coordinate $\zeta = w^{-1}z \in \mathbb{H}$. Writing $q = w/|w|$, the section and orbit coordinates are

$$(t, q, \zeta) \longmapsto \left(t, \frac{\sqrt{1-t^2}}{\sqrt{1+|\zeta|^2}} q, \frac{\sqrt{1-t^2}}{\sqrt{1+|\zeta|^2}} q\zeta \right). \tag{4.7}$$

Write the full normal coordinate as $u = (t, v) \in \mathbb{R} \oplus \mathbb{H}$. Its quaternionic component is

$$v = q^{-1}z = \frac{\sqrt{1-t^2}\,\zeta}{\sqrt{1+|\zeta|^2}}, \qquad \zeta = \frac{v}{\sqrt{1-t^2-|v|^2}}. \tag{4.8}$$

The map $(t, \zeta) \mapsto (t, v)$ is a diffeomorphism from $(-1, 1) \times \mathbb{H}$ onto $\operatorname{Int} D^5(1)$. Its Jacobian is

$$\det \frac{\partial(t, v)}{\partial(t, \zeta)} = \frac{(1-t^2)^2}{(1+|\zeta|^2)^3} > 0.$$

The coordinate change is an orientation-preserving diffeomorphism, so it preserves compact support and extends to degree $+1$ on the one-point compactifications. The pullback of b is therefore the nonzero supported loop of gauge transformations. The corresponding equivariant map is $qb(t, q^{-1}z)q^{-1}$, as in (4.4). □

Proposition 4.3. *The sphere in* (4.6) *is the Kervaire sphere:*

$$\Phi(\mathcal{A}R) = \kappa. \tag{4.9}$$

Proof. Let K^9 denote the boundary of the plumbing of two tangent disc bundles of S^5, oriented so that $[K^9] = \kappa$. This is the nonzero class of $bP_{10} \cong \mathbb{Z}/2$ and hence is exotic [11, 21]. We identify the clutched sphere with this manifold using the real pullback construction of [18, Section 4]. Write $\mathbb{R}^6 = \mathbb{R}e_0 \oplus \mathbb{R}^5 \cong \mathbb{C}^3$, with $ie_0 = e_1 \in \mathbb{R}^5$. On $S^9 \subset \mathbb{R}^5 \oplus \mathbb{R}^5$ consider

$$f(x, y) = \exp_{e_0}(\pi\tau(y/|y|)x), \qquad \tau(v)w = 2\langle v, w\rangle v - w.$$

Here $\exp_{e_0}$ is the exponential map of the unit round S^5, and $\tau(v)$ is reflection in the line spanned by v. Although the formula uses $y/|y|$, the map extends smoothly across $y = 0$. With $a = |x|^2$, it can be written as

$$f(x, y) = \cos(\pi\sqrt{a})e_0 - \frac{\sin(\pi\sqrt{a})}{\sqrt{a}}x + \frac{2\sin(\pi\sqrt{a})}{\sqrt{a}(1-a)}\langle x, y\rangle y.$$

Each scalar coefficient extends smoothly to $0 \le a \le 1$: at $a = 0$ use the power series of $\sin(\pi\sqrt{a})/\sqrt{a}$, and at $a = 1$ use its simple zero. Thus f is smooth. Orthogonal transformations commute with both the reflection formula and the round exponential map, so f is $SO(5)$-equivariant. It has norm one where the exponential formula is defined, and hence everywhere by continuity.

Use the first-column projections $SO(6) \to S^5$, $A \mapsto Ae_0$, and $SU(3) \to S^5$ given by the same formula. Thus $E = f^*SO(6) = \{(x, y, A) : Ae_0 = f(x, y)\}$. The free $\star$-action is $h \star (x, y, A) = (hx, hy, hA)$, with $SO(5)$ fixing e_0. Sperança's plumbing identification gives $E/SO(5)_\star \cong K^9$. Passing from the orthogonal construction to its determinant-positive component does not change this quotient: every $O(5)$ orbit meets that component in precisely an $SO(5)$ orbit.

Set $P = f^*SU(3) \subset E$, where $SU(3)$ is viewed as a subgroup of $SO(6)$ through its real action on $\mathbb{C}^3$. The subgroup $SO(5)$ fixes e_0, and $SU(2)$ is embedded in the last two complex coordinates. Transitivity of $SU(3)$ on S^5, with stabilizer $SU(2)$ at e_0, gives

$$SO(6) = SO(5)SU(3), \qquad SO(5) \cap SU(3) = SU(2).$$

They give a diffeomorphism $SO(5) \times_{SU(2)} P \longrightarrow E$, $[h, p] \mapsto h \star p$. Here the associated space is the quotient of $SO(5) \times P$ by $(h, p) \sim (hk^{-1}, k \star p)$ for $k \in SU(2)$, so the displayed map is well-defined. Consequently

$$P/SU(2)_\star \cong E/SO(5)_\star \cong K^9. \tag{4.10}$$

The bundle $P \to S^9$ is a principal right $SU(2)$-bundle: the right action multiplies the last two columns, while leaving the first column fixed. It commutes with left multiplication in the restricted $\star$-action.

On $\mathbb{R}^5 = \mathbb{R}e_1 \oplus \mathbb{C}^2$, the restricted $SU(2)$ action is trivial on e_1 and is the defining complex representation on $\mathbb{C}^2$. To identify it with left quaternionic multiplication, write $q = a + jb$, with $a, b \in \mathbb{C}$, and use

$$A(q) = \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix}, \qquad I(z_1, z_2) = z_1 + jz_2.$$

Then $I(A(q)z) = qI(z)$. The real map I sends the complex-oriented basis to $(1, i, j, -k)$; its determinant is -1, so $I \oplus I$ preserves orientation. Thus the base representation is $\mathbb{R}^2 \oplus \mathbb{H}^2$, with the left action used in (4.1). Choose poles $(e_1, 0)$ and $(-e_1, 0)$, both mapped by f to $-e_0$. The same matrix

$$A_0 = \operatorname{diag}(-1, i, i) \in SU(3)$$

specifies a point in each polar fiber and commutes with the embedded $SU(2)$. At either pole, $h \star A_0 = A_0 \cdot h$. Choose a principal connection and average it under the $\star$-action. The averaged form is still a principal connection, because the two actions commute. Parallel transport from these two fiber points along radial geodesics gives equivariant sections s_i satisfying $h \star s_i(y) = s_i(\rho_\kappa(h)y)h$. Consequently the $\star$-action in both trivializations is $(y, v) \mapsto (\rho_\kappa(h)y, hv)$. If $s_N(y) = s_S(y)\gamma(y)$, these identities imply $\gamma(\rho_\kappa(h)y) = h\gamma(y)h^{-1}$. Thus the transition between the two trivializations is conjugation equivariant.

By lemmas 4.1 and 4.2, such transitions have at most two homotopy-sphere classes. The constant loop gives the standard sphere, whereas (4.10) gives the nonstandard Kervaire sphere. The latter therefore corresponds to the nonzero element of $\pi_1(\mathcal{G}_H)$. The map b represents that element as well, proving (4.9). □

4.2. **Construction of λ.** Set

$$V_m = \mathbb{R}^m \oplus \mathbb{H} \oplus \operatorname{Im}\mathbb{H}, \qquad \rho_m(q)(a,w,z) = (a, qw, qzq^{-1}), \qquad m = 1,2. \tag{4.11}$$

We use $m = 1$ to identify the exotic 8-sphere and $m = 2$ to construct $\lambda \in \Theta_9$. On $w \neq 0$, where the action is free, put

$$q = \frac{w}{|w|}, \qquad v = q^{-1}zq.$$

Define the coordinates ψ_m by

$$\psi_m(q,a,v) = \left(a, q\sqrt{1-|a|^2-|v|^2}, qvq^{-1}\right), \qquad (a,v) \in \operatorname{Int} D^{m+3}(1). \tag{4.12}$$

Let $c : D^{m+3}(1) \to S^3$ be smooth and equal to 1 on a boundary neighborhood, as in the support convention of section 3. Define

$$\beta_c(a,w,z) = qc(a, q^{-1}zq)q^{-1} \tag{4.13}$$

on $w \neq 0$ and set it equal to 1 near $w = 0$. Since c is constant on a boundary collar, the two formulas agree on an invariant neighborhood; thus β_c is globally smooth and conjugation equivariant. In these coordinates, J_{β_c} acts by right multiplication on the S^3 factor:

$$\psi_m^{-1} \circ J_{\beta_c} \circ \psi_m(q,a,v) = (qc(a,v), a, v). \tag{4.14}$$

Lemma 4.4 (Equivariant normalization)**.** *For $m = 1,2$, every smooth conjugation-equivariant map $\beta : S(V_m) \to S^3$, after multiplication by a possible central constant -1, is equivariantly homotopic through smooth maps to some β_c of (4.13). Neither operation changes the oriented homotopy-sphere class defined by J_β.*

Proof. The set of points with nontrivial stabilizer is $F = \{w = 0\} = S(\mathbb{R}^m \oplus \operatorname{Im}\mathbb{H})$. At a point $(a,0,z)$ with $z \neq 0$ the stabilizer is the circle fixing the axis z, so equivariance forces

$$\beta(a,0,z) = \cos\vartheta(a) + \sin\vartheta(a)\frac{z}{|z|}$$

for a continuous circle-valued phase on D^m. A real lift exists because D^m is simply connected. At $z = 0$ the value is central. We first contract the phase continuously and then smooth the resulting equivariant homotopy relative to its prescribed data. For $m = 2$ the boundary circle is connected; after multiplication by -1 if necessary, the boundary value is 1, and the phase lifts to a real-valued function vanishing on the boundary. For $m = 1$, use the continuous section

$$(a,v) \longmapsto \left(a, \sqrt{1-a^2-|v|^2}, v\right), \qquad (a,v) \in D^4.$$

Composing with β extends $\beta|_F$ continuously over D^4. Replace its real phase by a homotopic smooth function with the same endpoint values in $\pi\mathbb{Z}$, constant near those endpoints. The round volume form of S^3 in these coordinates is $\sin^2\vartheta\, d\vartheta \wedge dV_{S^2}$. Since $\operatorname{vol}(S^2) = 4\pi$ and $\operatorname{vol}(S^3) = 2\pi^2$, the degree, up to the fixed boundary orientation sign, is

$$\frac{2}{\pi}\int_{-1}^{1} \sin^2\vartheta(a)\vartheta'(a)\, da = \frac{\vartheta(1) - \vartheta(-1)}{\pi}.$$

The last equality follows from $\int \sin^2 t\, dt = t/2 - \sin(2t)/4$ and the endpoint values. The extension over D^4 forces this degree to vanish, so the endpoint values of the lift agree. After a central normalization and subtracting a constant multiple of 2π, the phase vanishes at both endpoints.

Contracting the phase makes $\beta|_F$ equal to 1. The invariant submanifold F has an equivariant tubular neighborhood, so the pair $(S(V_m), F)$ has the equivariant homotopy extension property. Extend the contraction continuously over the sphere. At its final time the map is uniformly close to 1 on an invariant neighborhood of F. In that neighborhood use the equivariant logarithm at 1 and an invariant cutoff to make the final map exactly 1 on a smaller neighborhood. Arrange the homotopy to be stationary near its two time endpoints.

Apply the relative equivariant smoothing procedure following lemma 4.1, fixing the original smooth endpoint and the final constant neighborhood. This gives a smooth equivariant homotopy ending at a map equal to 1 near F. Its restriction to the free section is a smooth map c equal to 1 near ∂D^{m+3}.

By lemma 2.2, an equivariant homotopy gives an isotopy of the maps J_β. Multiplication by -1 composes J_β with a fixed orientation-preserving linear map that extends over the disc. Hence neither change alters the oriented sphere class. □

A homomorphism to $\mathrm{Diff}_\partial(D^6)$. We construct a homomorphism that relates the maps J_{β_c} to the Gromoll construction. Identify D^3 with the unit disc in $\mathrm{Im}\,\mathbb{H}$. For $0 < \sigma < 1$, let

$$\mathcal{G}_c = \{c \in C^\infty(D^3, S^3) : c(v) = 1 \text{ for } |v| \geq \sigma\}, \qquad \mathcal{D} = \mathrm{Diff}_\partial(D^6),$$

with pointwise multiplication on $\mathcal{G}_c$ and the C^∞ topologies. Relative smoothing and radial compression show that inclusion in $\mathrm{Map}_*(D^3/\partial D^3, S^3)$ is a weak equivalence of topological groups. In particular, $\pi_r\mathcal{G}_c \cong \pi_{r+3}(S^3)$ and $\pi_0\mathcal{G}_c \cong \mathbb{Z}$.

Use the representation $\rho_0(q)(w, z) = (qw, qzq^{-1})$ on $S(\mathbb{H}\oplus\mathrm{Im}\,\mathbb{H}) = S^6$. The region $w \neq 0$ has coordinates

$$\psi_0(q, v) = \Big(\sqrt{1 - |v|^2}\, q, qvq^{-1}\Big), \qquad q \in S^3, \quad v \in \mathrm{Int}\, D^3.$$

The inverse recovers $q = w/|w|$ and $v = q^{-1}zq$; in particular $|v| = |z|$. Define the sphere diffeomorphism $\mathcal{T}(c)$ by

$$\mathcal{T}(c)\Big(\psi_0(q, v)\Big) = \psi_0(qc(v)^{-1}, v), \tag{4.15}$$

and extend it by the identity near $w = 0$. This extension is smooth because $c = 1$ for $|v| \geq \sigma$. All these maps fix the same open set $\{|z| > \sigma\}$. Choose an open round cap C whose closure lies in that set, and fix an orientation-preserving parametrization $j : D^6 \to S^6 \setminus C$. Every $\mathcal{T}(c)$ is the identity on a neighborhood of the closed cap, hence preserves its complement and is the identity near its boundary. Restriction and conjugation by j therefore define $\mathcal{T} : \mathcal{G}_c \to \mathcal{D}$; we use the same symbol for the sphere map and this disc map.

For $r = 1, 2$ define

$$\Lambda_r : \pi_r\,\mathrm{Diff}_\partial(D^6) \longrightarrow \Theta_{r+7} \tag{4.16}$$

as follows. Choose a smooth representative family f_a, $a \in D^r$, such that $f_a(z) = z$ near the entire boundary of $D^r \times D^6$. Form $\widehat{f}(a, z) = (a, f_a(z))$ and round the corners within its identity collar. Extend the resulting D^{r+6}-diffeomorphism by the identity to S^{r+6}, and clutch two $(r + 7)$-discs. This is the Gromoll homomorphism with the clutching convention fixed in Section 2 [4, Section 2.1].

Lemma 4.5 (The Gromoll construction for $\mathcal{T}$)**.** *The map $\mathcal{T}$ is a continuous group homomorphism. Let $r = 1$ or 2, and let $c : D^r \times D^3 \to S^3$ be a smooth family*

representing a class in $\pi_r\mathcal{G}_c$, equal to 1 near the product boundary. After compressing its support into the unit normal disc as below, the associated attaching map satisfies

$$[\Sigma_{J_{\beta_c}}] = -\Lambda_r(\mathcal{T}_*[c]) = \Lambda_r(\mathcal{T}_*[c]) \in \Theta_{r+7}.$$

Here Θ_8 and Θ_9 have exponent two.

Proof. In the fixed coordinates ψ_0, applying $\mathcal{T}(d)$ and then $\mathcal{T}(c)$ sends

$$(q, v) \longmapsto (qd(v)^{-1}c(v)^{-1}, v) = (q(c(v)d(v))^{-1}, v).$$

Thus $\mathcal{T}(c) \circ \mathcal{T}(d) = \mathcal{T}(cd)$, both on S^6 and on the fixed disc. The inverse is $\mathcal{T}(c^{-1})$. The common identity neighborhood ensures smooth extension, and the coordinate formula gives continuity in the C^∞ topology.

For the family assertion, use orientation-preserving relative coordinate compressions to arrange

$$\operatorname{supp}(c-1) \subset \{|a| < a_*,\ |v| < d_*\}, \qquad 0 < a_* < 1, \quad 0 < d_* < \sigma\sqrt{1-a_*^2}.$$

These compressions are isotopic to the identity and preserve the relative class. Extend c by 1 outside the product disc. The support condition implies $a_*^2 + d_*^2 < 1$, so restricting this extension to the unit normal disc gives the map c used to define J_{β_c}. Write a point of S^{r+6} as $(a, \sqrt{1-|a|^2}\, x)$, with $x \in S^6$ and $|a| < 1$. In the coordinates ψ_0, its induced family is

$$x \longmapsto \mathcal{T}(c_a)^{-1}(x), \qquad c_a(v) = c(a, \sqrt{1-|a|^2}\, v).$$

For $|a| \le a_*$ the normal support of c_a lies in the fixed disc of radius $d_*/\sqrt{1-a_*^2} < \sigma$; for $|a| \ge a_*$ the map is constant. Thus every c_a lies in the same group $\mathcal{G}_c$ and every $\mathcal{T}(c_a)$ fixes the chosen cap. The rescaling is a relative homotopy: replace $\sqrt{1-|a|^2}$ by $1 - t + t\sqrt{1-|a|^2}$, $0 \le t \le 1$. The same support estimate holds throughout. The family is constant near $|a| = 1$ and near the product boundary, so it remains smooth after rounding the corners.

Choose

$$a_* < a_1 < 1. \tag{4.17}$$

Use the fixed parametrization $j : D^6 \to S^6 \setminus C$ chosen above. The map

$$E : D^r(a_1) \times D^6 \longrightarrow S^{r+6}, \qquad E(a, x) = \left(a, \sqrt{1-|a|^2}\, j(x)\right) \tag{4.18}$$

is a smooth embedding, including its corners: the scale is at least $\sqrt{1-a_1^2} > 0$, and the inverse on its image recovers a and then $j(x)$ by division by that scale.

The entire nontrivial support lies in the interior of this smaller product. On its parameter boundary collar, $|a| > a_*$ and the family is constant; near the D^6 boundary, every $\mathcal{T}(c_a)$ fixes the common cap. Thus in the chart (4.18) the sphere map is exactly

$$(a, x) \longmapsto (a, \mathcal{T}(c_a)^{-1}(x)),$$

with a common identity collar on the whole product boundary. Round the corners within that collar, and rescale the parameter disc with the orientation transported by the rescaling. This gives the defining Gromoll representative [21, Section 1.1], equivalently the cubical model with fixed boundary collars in [4, Section 2.1]. Extension by the identity outside the chart recovers the original sphere map, and inversion accounts for the minus sign in the statement. □

Precomposition with the Hopf map. For a topological group H, precomposition with the Hopf map $\eta_2 : S^3 \to S^2$ defines the following operation on pointed homotopy sets:

$$\mathfrak{h}_H : \pi_1 H \cong \pi_2 BH \xrightarrow{\circ\eta_2} \pi_3 BH \cong \pi_2 H. \tag{4.19}$$

It is natural under continuous group homomorphisms. The universal principal bundle gives the comparison $B\mathcal{G}_c \simeq \mathrm{Map}_*(S^3, BS^3)$. Applying $\mathrm{Map}_*(S^3, -)$ to the universal principal bundle $S^3 \to ES^3 \to BS^3$ gives a fibration with contractible total space and fiber $\mathrm{Map}_*(S^3, S^3)$. Its base is connected since $\pi_3(BS^3) = \pi_2(S^3) = 0$. Relative smoothing and the fixed-collar compression identify this fiber weakly with $\mathcal{G}_c$. Under adjunction, precomposition by η_2 becomes precomposition by $\Sigma^3\eta_2 = \eta_5$ in BS^3. To pass to the fiber, write $\partial_j : \pi_j(BS^3) \to \pi_{j-1}(S^3)$ for the connecting isomorphism. Since $\eta_5 = \Sigma\eta_4$, loop adjunction and naturality of the map $\Omega BS^3 \to S^3$ give

$$\partial_6(v \circ \eta_5) = (\partial_5 v) \circ \eta_4, \qquad v \in \pi_5(BS^3).$$

Taking $\partial_5 v = \eta_3$ therefore gives

$$\mathfrak{h}_{\mathcal{G}_c}(\eta_3) = \eta_3\eta_4. \tag{4.20}$$

Lemma 4.6 (Gromoll–Morlet compatibility)**.** *For every $x \in \pi_1\mathcal{D}$,*

$$p_9\Lambda_2(\mathfrak{h}_{\mathcal{D}}x) = \big(p_8\Lambda_1(x)\big)\eta \quad \textit{in } \operatorname{Coker} J_9. \tag{4.21}$$

Proof. Let $O_j = O(j)$. We use the simplicial groups PL_j and TOP_j of piecewise-linear and topological automorphisms of $\mathbb{R}^j$ fixing the origin, in the smoothing models of [16]. The notation PL_j/O_j denotes the homotopy fiber of $BO_j \to BPL_j$. For a based map $f : X \to Y$, its homotopy fiber is modeled by pairs (x, γ) with a path γ from $f(x)$ to the basepoint of Y. This homotopy fiber is represented by PD_j/O_j, with PD_j the piecewise-differentiable intermediary in [16, Sections 5.1, 8.1 and 9.2]. The map $PD_j \to TOP_j$ supplies its natural comparison with TOP_j/O_j. Write PL/O for the stable smoothing space and put $X_6 = PL_6/O_6$, $Y_6 = TOP_6/O_6$, with these conventions throughout. Use the smoothing comparison $M_6 : \pi_i\mathcal{D} \to \pi_{i+7}X_6$ obtained from [16, Section 9.2]. For a smooth relative diffeomorphism f of D^6, both the PD and TOP framed models assign the derivative Df and the path of germs

$$H^f_{z,t}(v) = \begin{cases} (f(z+tv) - f(z))/t, & t > 0, \\ Df_z(v), & t = 0. \end{cases}$$

The forgetful map from PD to TOP preserves these data and the fibers over the identity. It induces the natural map between the two comparison diagrams in [16, Section 9.2, after equation (9.4)]; the displayed path is equation (8.4) there. For relative diffeomorphisms f, g, the composition identity is

$$H^{f\circ g}_{z,t} = H^f_{g(z),t} \circ H^g_{z,t}.$$

It holds for $t > 0$ by substitution and at $t = 0$ by the chain rule. Thus the TOP map preserves composition, as in Lemma 8.1. The fiber inclusion and Proposition A.1 then identify this comparison as an E_1 equivalence $\mathcal{D} \simeq \Omega^7 Y_6$, where E_1 denotes multiplication associative up to coherent homotopy; see the proof of Theorem 9.1 in that paper. Delooping gives $B\mathcal{D} \simeq \Omega^6_0 Y_6$, with subscript 0 denoting the component of the constant map. By adjunction, precomposition by η_2 corresponds to precomposition by $\Sigma^6\eta_2 = \eta_8$ in Y_6.

The natural comparison lies in the fiber sequence

$$X_6 \xrightarrow{j} Y_6 \longrightarrow TOP_6/PL_6 \simeq K(\mathbb{Z}/2, 3)$$

from [16, equation (2.3) and Section 9.2]. Here $K(\mathbb{Z}/2, 3)$ denotes the Eilenberg–MacLane space whose only nonzero homotopy group is $\pi_3 = \mathbb{Z}/2$. In particular its homotopy groups in degrees 8, 9, and 10 vanish. The long exact sequence therefore makes j_* an isomorphism on π_8 and π_9. The preceding map between comparison diagrams identifies j_*M_6 with the TOP comparison. It follows that $j_*M_6(\mathfrak{h}_{\mathcal{D}}x) = (j_*M_6x) \circ \eta_8$. Postcomposition with j commutes with precomposition by η_8. Since j_* is injective on π_9, it follows that

$$M_6(\mathfrak{h}_{\mathcal{D}}x) = (M_6x) \circ \eta_8. \tag{4.22}$$

For the geometric Gromoll map of (4.16), lemma C.1 identifies this smoothing comparison with stabilization. Let $\Psi_j : \Theta_j \to \pi_j(PL/O)$ be the smoothing-class isomorphism and $S : X_6 \to PL/O$ the stabilization map. Then

$$\Psi_{r+7}\Lambda_r = S_*M_6, \qquad r = 1, 2. \tag{4.23}$$

The smoothing class of the glued sphere is identified in [4, Lemmas 2.5 and 2.7].

It remains to identify the normal invariant of the clutched sphere. Let G be the stable monoid of based homotopy self-equivalences of spheres, based at the identity. The canonical maps $BO \to BPL \to BG$ forget a stable vector-bundle structure first to a piecewise-linear bundle and then to its underlying stable spherical fibration. As with the smoothing spaces above, we use homotopy-fiber models:

$$PL/O = \operatorname{hofiber}(BO \to BPL), \qquad G/O = \operatorname{hofiber}(BO \to BG).$$

The identity on BO and the map $BPL \to BG$ induce the normal-invariant map $\mathfrak{n} : PL/O \to G/O$. Under the canonical identification $\pi_j(G) \cong \pi_j^s$ for $j > 0$, the map $O \to G$ induces J_j. The homotopy-fiber sequence $O \to G \to G/O$ therefore gives, for $j = 8, 9$, an injection

$$\jmath_j : \operatorname{Coker} J_j \hookrightarrow \pi_j(G/O)$$

whose image is the torsion subgroup of $\pi_j(G/O)$ [4, Section 2.2]. The group $\pi_j(PL/O) \cong \Theta_j$ is finite, so the image of $\mathfrak{n}_*$ lies in this subgroup. We may therefore define

$$P_j = \jmath_j^{-1} \circ \mathfrak{n}_* : \pi_j(PL/O) \longrightarrow \operatorname{Coker} J_j,$$

using the inverse of $\jmath_j$ on its image. By [4, Proposition 2.10], $P_j\Psi_j = p_j$.

Precomposition by $\eta_8 : S^9 \to S^8$ commutes with S and $\mathfrak{n}$. Its descent to the two J-cokernels can be checked directly:

$$\operatorname{Im} J_8 = \langle \bar{\nu} + \epsilon \rangle, \qquad (\bar{\nu} + \epsilon)\eta = \eta\bar{\nu} + \eta\epsilon \in \operatorname{Im} J_9. \tag{4.24}$$

The image descriptions are given in [21, Appendix C, Table 4]. Since $|\bar{\nu}| = |\epsilon| = 8$, graded commutativity has sign $+1$. Thus multiplication by η descends to $\operatorname{Coker} J_8 \to \operatorname{Coker} J_9$. This is the required special case of [21, Proposition 2.4], with $(n, i, t) =$

$(6,1,1)$. Consequently

$$\begin{array}{ccc} \pi_1\mathcal{D} & \xrightarrow{\mathfrak{h}_{\mathcal{D}}} & \pi_2\mathcal{D} \\ {\scriptstyle M_6}\downarrow & & \downarrow{\scriptstyle M_6} \\ \pi_8 X_6 & \xrightarrow{\circ\eta_8} & \pi_9 X_6 \\ {\scriptstyle P_8 S_*}\downarrow & & \downarrow{\scriptstyle P_9 S_*} \\ \operatorname{Coker} J_8 & \xrightarrow{\cdot\eta} & \operatorname{Coker} J_9 \end{array}$$

commutes. By (4.23), its vertical composites are $p_8\Lambda_1$ and $p_9\Lambda_2$. This proves the lemma. □

Lemma 4.7 (Identification of the exotic 8-sphere)**.** *For the nonzero class $\eta_3 \in \pi_4(S^3) = \pi_1\mathcal{G}_{\mathrm{c}}$,*

$$\Lambda_1(\mathcal{T}_*\eta_3) = \Sigma^8_{[\epsilon]}, \qquad p_8(\Sigma^8_{[\epsilon]}) = [\epsilon]. \tag{4.25}$$

Proof. Sperança's pullback is defined by

$$f_8(\ell, w, z) = \frac{(\ell + wi\overline{w},\ z)}{\sqrt{\ell^2 + |w|^4 + |z|^2}} : S(\mathbb{R}\oplus\mathbb{H}\oplus\mathbb{H}) \longrightarrow S^7,$$

using the first-column projection $Sp(2) \to S^7$. On the domain,

$$\ell^2 + |w|^4 + |z|^2 = 1 - |w|^2 + |w|^4 \geq \tfrac{3}{4}.$$

Since $wi\overline{w}$ is imaginary, the squared norm of the numerator is exactly the same expression. Hence f_8 is a globally smooth sphere-valued map. Let $P_8 = f_8^* Sp(2)$ be the pullback bundle. Write a point as $(x, (v_1, v_2))$, where $x = (\ell, w, z)$, $v_1 = f_8(x)$, and $(v_1, v_2) \in Sp(2)$. Its principal right action is $(v_1, v_2)\cdot h = (v_1, v_2 h)$. The commuting $\star$-action is

$$q \star \Big((\ell, w, z), (v_1, v_2)\Big) = \Big((\ell, qw, qzq^{-1}), (qv_1q^{-1}, qv_2)\Big).$$

Quaternionic multiplication acts componentwise on the columns and preserves their orthonormality. Equivariance of f_8 therefore makes this an action on P_8. It is free: $qv_2 = v_2$ for a nonzero column forces $q = 1$. Sperança identifies $P_8/S^3_\star$ with the unique exotic 8-sphere [18, Section 2.1 and Theorem 1]. At the two poles $(\pm 1, 0, 0)$, the first columns are $(\pm 1, 0)$. Choose $(0, 1)$ for the second column in each fiber. At these chosen fiber points, the action gives $(0, q) = (0, 1)q$, agreeing with principal right multiplication by q. Equivariant radial parallel transport therefore gives two trivializations whose transition on the equator is conjugation equivariant. Writing $z = a + z_0$ with $a = \operatorname{Re} z$ and $z_0 \in \operatorname{Im}\mathbb{H}$ identifies that equatorial representation with $V_1 = \mathbb{R}\oplus\mathbb{H}\oplus\operatorname{Im}\mathbb{H}$. By lemma 4.4, its transition is represented by one of the two elements of $\pi_4(S^3)$. The zero element has a null homotopy equal to 1 on a common boundary neighborhood, which gives an isotopy of the attaching map to the identity and hence the standard sphere. Since this quotient is exotic, the resulting map c represents η_3. The conclusion follows from lemma 4.5.

□

For $m = 2$, put

$$V_\lambda = V_2 = \mathbb{R}^2 \oplus \mathbb{H} \oplus \operatorname{Im}\mathbb{H}, \qquad \rho_\lambda(q)(a, w, z) = (a, qw, qzq^{-1}), \tag{4.26}$$

and, for $q \in S^3$ and $(a, z) \in D^5(c_1) \subset \mathbb{R}^2 \oplus \operatorname{Im}\mathbb{H}$, define

$$e_\lambda(q, a, z) = \Big(a, \sqrt{1 - |a|^2 - |z|^2}\, q, qzq^{-1}\Big). \tag{4.27}$$

Use the orthogonal permutation $\Pi_\lambda : \mathbb{H} \oplus (\mathbb{R}^2 \oplus \operatorname{Im}\mathbb{H}) \to V_\lambda$ given by $\Pi_\lambda(w, (a, v)) = (a, w, v)$, and put $T_{\tau_2}(q, u) = (q, \tau_2(q)u)$. The permutation preserves orientation and extends linearly over the 9-disc. The explicit normal-framing homotopy of lemma 3.4 gives

$$e_\lambda = \Pi_\lambda \circ e_0 \circ T_{\tau_2} \simeq \Pi_\lambda \circ e_0 \circ T_\tau^2 \tag{4.28}$$

through parametrized tubular embeddings with fixed core. Conjugation by Π_λ and isotopy extension preserve the clutched sphere class. Taking $c = b$ in (4.13), set

$$\beta_\lambda(e_\lambda(q, u)) = qb(u)q^{-1}. \tag{4.29}$$

In the coordinates e_λ, the attaching diffeomorphism is R, so

$$[\Sigma_{J_{\beta_\lambda}}] = \Phi(\mathcal{A}^2 R). \tag{4.30}$$

Define

$$\lambda := \Phi(\mathcal{A}^2 R). \tag{4.31}$$

Proposition 4.8. *The class λ satisfies*

$$p_9(\lambda) = [\eta\epsilon] = [\nu^3] \neq 0, \qquad \alpha(\lambda) = 0.$$

In particular, $\lambda \notin bP_{10}$.

Proof. By naturality of (4.19), (4.20), and the fact that $\mathcal{T}$ is a group homomorphism,

$$\mathcal{T}_*(\eta_3\eta_4) = \mathfrak{h}_{\mathcal{D}}(\mathcal{T}_*\eta_3).$$

Combining lemmas 4.5 and 4.6 with (4.25) gives

$$p_9(\lambda) = p_9\Lambda_2\Big(\mathfrak{h}_{\mathcal{D}}(\mathcal{T}_*\eta_3)\Big) = [\epsilon]\eta = [\epsilon\eta].$$

Stable homotopy is graded commutative, so $\epsilon\eta = \eta\epsilon$. Relation C.2 and Table 4 of [21] give $[\eta\epsilon] = [\nu^3] \neq 0$ in $\operatorname{Coker} J_9$.

The factorization (2.3) now gives

$$\alpha(\lambda) = \bar{\alpha}_9([\epsilon\eta]) = \alpha_8^{\mathrm{fr}}(\epsilon)\alpha_1^{\mathrm{fr}}(\eta) = 0.$$

□

In the Gromoll filtration, the subgroup $\Gamma_3^9 = \operatorname{im}(\Lambda_2) \subset \Theta_9$ contains λ. Wang proves that $p_9 : \Gamma_3^9 \to \operatorname{Coker} J_9$ is split surjective and constructs a class with image $[\epsilon\eta]$ [21, Corollary 3.2 and Section 4.1.2].

5. The class $\kappa + \lambda$

We use lemma 3.7 and quaternionic inversion to compute $\Phi(\mathcal{A}L)$.

Proposition 5.1 (A relation in $\mathcal{E}$)**.** *The class $R \in \mathcal{E}$ satisfies*

$$\mathcal{A}^2 R = R\,\iota(\lambda). \tag{5.1}$$

Proof. By lemma 3.7, $(\mathcal{A}^2 R)\mathcal{N} = R\mathcal{N}$. Thus $\mathcal{A}^2 R = R\iota(w)$ for a unique $w \in \Theta_9$. Applying Φ, and using $\Phi(R) = 0$ and $\Phi\iota = \mathrm{id}$, identifies $w = \lambda$. □

Lemma 5.2 (Quaternionic inversion)**.** *For a smooth based map $\tau : S^3 \to SO(5)$, write $T_\tau(q,u) = (q, \tau(q)u)$. Let $\widehat{R}_{\tau,b}$ and $\widehat{L}_{\tau,b}$ be the diffeomorphisms of S^8 obtained by transporting R_b and L_b through the embedding $e_0 T_\tau$ and extending by the identity outside its image. Here $R_b(q,u) = (qb(u),u)$ and $L_b(q,u) = (b(u)q,u)$. Set $\tau^\vee(q) = \tau(q^{-1})$ and $\mathcal{I}(w,u) = (\overline{w},u)$ on $S(\mathbb{H}\oplus\mathbb{R}^5)$. Then*

$$\mathcal{I}\widehat{R}_{\tau,b}\mathcal{I}^{-1} = \widehat{L}_{\tau^\vee,b^{-1}}. \tag{5.2}$$

In particular, with the fixed generator b and for every $m \in \mathbb{Z}$,

$$\Phi(\mathcal{A}^m R) = \Phi(\mathcal{A}^{-m} L). \tag{5.3}$$

Proof. The ambient map $\mathcal{I}$ preserves the image of the standard embedding e_0, since quaternionic conjugation preserves $|w|$. In the coordinates e_0, it acts as $\jmath(q,u) = (q^{-1},u)$. It therefore suffices to compute the conjugation on T. Direct composition gives

$$\jmath T_\tau \jmath = T_{\tau^\vee}, \qquad \jmath R_b \jmath = L_{b^{-1}}.$$

Conjugating the product $T_\tau R_b T_\tau^{-1}$ proves (5.2) on $e_0(T)$ and its complement, where both sides are the identity. The common identity collar extends this equality smoothly to S^8.

The map $q \mapsto q^{-1}$ has degree -1 on S^3, hence $[\tau^\vee] = -[\tau]$ in $\pi_3 SO(5)$. Moreover, b^{-1} is homotopic to b through maps equal to 1 near the boundary, since $\pi_5(S^3)$ has order two. The ambient map $\mathcal{I}$ reverses orientation and extends linearly over D^9; conjugation by it negates the clutched sphere class. Negation is trivial in Θ_9. Taking $[\tau] = m[\tau_1]$ gives (5.3). □

Theorem 5.3 (Identification of $\kappa+\lambda$)**.** *The class $\mathcal{A}L$ satisfies*

$$\Phi(\mathcal{A}L) = \kappa + \lambda. \tag{5.4}$$

Proof. Apply $\mathcal{A}^{-1}$ to (5.1). Since $\mathcal{A}^{-1}$ fixes $\mathcal{N}$ pointwise, this gives

$$\mathcal{A}R = (\mathcal{A}^{-1}R)\iota(\lambda).$$

Hence $\Phi(\mathcal{A}^{-1}R) = \kappa - \lambda = \kappa + \lambda$. Quaternionic inversion, (5.3) with $m=-1$, now gives $\Phi(\mathcal{A}L) = \Phi(\mathcal{A}^{-1}R)$. □

5.1. **An attaching map for $\kappa+\lambda$.** Return to the representation V_κ in (4.1). On the open set $w \neq 0$, set $q = w/|w|$ and define

$$b_\kappa(t,w,z) = b(t, q^{-1}z). \tag{5.5}$$

If $b(t,q^{-1}z) \neq 1$, then $t^2 + |z|^2 < c_0^2$, and hence $|w|^2 > 1 - c_0^2$. Thus (5.5) equals 1 on a neighborhood of $\{w=0\}$ and extends smoothly by that value. It satisfies

$$b_\kappa(\rho_\kappa(h)y) = b_\kappa(y). \tag{5.6}$$

Define

$$L_{b_\kappa}(y) = \rho_\kappa(b_\kappa(y))y. \tag{5.7}$$

By lemma 2.2, this is a diffeomorphism with inverse $L_{b_\kappa^{-1}}$. In the coordinates (4.2),

$$e_\kappa^{-1} \circ L_{b_\kappa} \circ e_\kappa(q,t,z) = (b(t,z)q,t,z),$$

so, relative to the standard embedding e_0, it is the class $\mathcal{A}L$. Thus theorem 5.3 identifies the sphere obtained by gluing along L_{b_κ}.

Corollary 5.4. *The homotopy sphere*

$$\Sigma_{\kappa+\lambda} := D^9_N \cup_{L_{b_\kappa}} D^9_S$$

represents $\kappa + \lambda \in \Theta_9$.

6. Positively curved metrics on the two discs

We construct compatible positively curved metrics on two discs for the conjugation-equivariant transitions in [9, 6, 5], and for the invariant transition that gives L_{b_κ} in corollary 5.4.

For a free isometric $\star$-action on (P, G), let $\varpi : P \to P/S^3_\star$ be the quotient map. A vector is *horizontal for* ϖ if it lies in $(\ker d\varpi)^{\perp_G}$. The quotient metric $\bar{G}$ is defined by

$$\langle d\varpi_p X, d\varpi_p Y\rangle_{\bar{G}} = \langle X, Y\rangle_G \quad \text{for } X, Y \in (\ker d\varpi_p)^{\perp_G}.$$

The isometric action makes this definition independent of the point chosen in the orbit, and ϖ is a Riemannian submersion. Its horizontal space generally differs from the horizontal space $\ker\omega$ of the principal connection. For a quotient plane Π and its horizontal lift $\tilde{\Pi}$, O'Neill's formula gives $\sec_{\bar{G}}(\Pi) \geq \sec_G(\tilde{\Pi})$ [14].

We use the following specialization of the smooth gluing theorem.

Theorem 6.1 (Reiser–Wraith)**.** *Let (M_N, g_N) and (M_S, g_S) be compact smooth Riemannian n-manifolds with boundary, $n \geq 3$, whose sectional curvatures are strictly positive, including at the boundary. Let $\phi : \partial M_N \to \partial M_S$ be a boundary isometry, and write $B_i(Y, Z) = \langle \nabla_Y n_i, Z\rangle_{g_i}$ for the second fundamental form with outward unit normal n_i. If $B_N + \phi^* B_S \geq 0$, then $M_N \cup_\phi M_S$ admits a smooth metric of positive sectional curvature agreeing with g_N, g_S outside any prescribed neighborhood of the interface.*

Proof. This is [15, Theorem A(i), $k = 1$]. There Ric_k is a sum of k sectional curvatures through a unit vector, so $\mathrm{Ric}_1 > 0$ is precisely positive sectional curvature. Their inward-normal convention $\mathrm{II}(Y, Z) = -\langle \nabla_Y n_{\mathrm{in}}, Z\rangle_{g_i}$ agrees with the outward-normal form B_i above. □

Corollary 6.2 (Equivariant gluing)**.** *Under the hypotheses of theorem 6.1, suppose a compact Lie group acts isometrically on both pieces and ϕ is equivariant. Then the glued metric can be chosen invariant.*

Proof. The inward normal exponential maps give equivariant collars. In each collar the group acts on the boundary coordinate and fixes the distance coordinate t. Write one collar metric as $dt^2 + h_t$ and choose a smooth radial cutoff χ_0 supported in that collar, equal to one near $t = 0$. For a constant $\sigma > 0$, replace its slice metrics by

$$h^\sigma_t = h_t - 2\sigma t\chi_0(t)h_0.$$

This invariant perturbation fixes the boundary metric and changes the outward second fundamental form from $-\frac{1}{2}h'_0$ to $-\frac{1}{2}h'_0 + \sigma h_0$. It tends to zero in C^2 as $\sigma \to 0$, with the collar and cutoff fixed. Compactness and the strict sectional-curvature bound therefore preserve positivity for small σ, while the sum of boundary forms becomes positive definite.

The cubic interpolation in [15, Section 2.1, equation (1)] is a linear combination, with radial scalar coefficients, of the two slice metrics and their radial derivatives. It is consequently invariant. For the smoothing in [15, Section 2.2], use the same

radial mollifier and radial cutoff for all tensor components. Tangential coordinate changes are independent of t, so this defines smoothing of the slice tensor itself and commutes with pullback by the group. The curvature estimates in that construction allow common widths on the compact boundary. The resulting invariant smooth metric has positive sectional curvature and agrees with the original metrics outside the prescribed neighborhood. □

6.1. **Connection metrics.** We use the sign convention for R^g fixed in the introduction. In the curvature calculations, R denotes the curvature tensor of the total-space metric, and R^B denotes that of the base metric. We first state the quantitative connection-metric criterion used on the southern disc, following [9, Proposition 3.1]. The curvature calculation is given in Appendix A.

Theorem 6.3 (Positive connection metrics)**.** *Let*

$$\pi : P \longrightarrow (B, g_B)$$

be a principal right S^3-bundle over a compact Riemannian manifold, possibly with boundary, and assume $\sec_{g_B} \geq \kappa_0 > 0$. Put $\mathfrak{g} = \operatorname{Im}\mathbb{H}$, and let $Q_{\mathfrak{g}}$ be its inner product for which the induced bi-invariant metric on S^3 has sectional curvature one; we use h_{S^3} for that metric on the fiber. Let ω be a smooth principal connection. Regard its curvature Ω as an adjoint-bundle-valued two-form on B; the same symbol denotes its corresponding equivariant horizontal form on P. Put

$$M_0 = \sup_B |\Omega|\,, \qquad M_1 = \sup_B |D\Omega|\,.$$

Here D uses the base Levi–Civita connection and the induced adjoint connection. In base and Lie-algebra orthonormal frames we use $|\Omega|^2 = \sum_{i<j,a}(\Omega^a_{ij})^2$ and $|D\Omega|^2 = \sum_{k,i<j,a}((D_k\Omega)^a_{ij})^2$. Set $C_0 = 16$ and choose

$$\Lambda \geq 16C_0M_0^2 + 32C_0^2\kappa_0^{-1}(M_1+1)^2 + 8. \tag{6.1}$$

For $\varphi \in C^\infty(B)$, its Hessian is $\operatorname{Hess}_{g_B}\varphi(X,Y) = X(Y\varphi) - (\nabla^B_X Y)\varphi$. Suppose $-\operatorname{Hess}_{g_B}\varphi(X,X) \geq \Lambda\langle X,X\rangle_{g_B}$ for every base tangent vector X. Then the connection metric

$$G_\varepsilon = \pi^* g_B + r_\varepsilon^2 Q_{\mathfrak{g}}(\omega,\omega), \qquad r_\varepsilon^2 = \varepsilon e^{\varepsilon\varphi} \tag{6.2}$$

has strictly positive sectional curvature whenever $0 < \varepsilon \leq \varepsilon_{\mathrm{conn}}$, where the positive number $\varepsilon_{\mathrm{conn}}$ is given explicitly in (A.14). More precisely,

$$\sec_{G_\varepsilon} \geq \min\left\{\frac{\kappa_0}{2}, \frac{\varepsilon\Lambda}{8}, \frac{1}{4\varepsilon}\right\} > 0. \tag{6.3}$$

The functions in (6.2) are pulled back from B.

Proof. The curvature computation and simultaneous estimates in Appendix A give (A.17) under the stated hypotheses. To deduce the sectional bound, take independent tangent vectors $P = X + U$ and $Q = Y + V$, split into horizontal and vertical parts with respect to ω. With norms induced by G_ε, put

$$h = |X \wedge Y|, \qquad m = |X \otimes V - Y \otimes U|, \qquad k = |U \wedge V|.$$

These are the norms of the three orthogonal components of $P \wedge Q$, so $|P \wedge Q|^2 = h^2 + m^2 + k^2$. The cited estimate states

$$\langle R(P,Q)Q, P\rangle_{G_\varepsilon} \geq \frac{\kappa_0}{2}h^2 + \frac{\varepsilon\Lambda}{8}m^2 + \frac{1}{2r_\varepsilon^2}k^2.$$

The same scale choice gives $r_\varepsilon^2 \le 2\varepsilon$. Dividing by the positive squared area $|P \wedge Q|^2$ proves (6.3). □

6.2. **Two S^3-actions on the bundle.** Let V be an orthogonal real representation of S^3 of dimension $n \ge 3$, and let

$$\rho : S^3 \longrightarrow O(V).$$

On the unit sphere $S(V)$ define the infinitesimal action map

$$K_y : \operatorname{Im}\mathbb{H} \longrightarrow T_y S(V), \qquad K_y\xi = \left.\frac{\mathrm{d}}{\mathrm{d}s}\right|_{s=0} \rho(e^{s\xi})y. \tag{6.4}$$

We use the unit round metric on $S(V)$ and the standard Euclidean metric on $\operatorname{Im}\mathbb{H}$, and fix

$$L_\rho = \sup_{y\in S(V)} \|K_y\|_{\mathrm{op}} < \infty. \tag{6.5}$$

Let U_N, U_S be the closed northern and southern hemispheres of $S(\mathbb{R}\oplus V)$, with common equator $S(V)$. The base action is trivial on $\mathbb{R}$ and is ρ on V. We use the equivariant disc charts

$$x \longmapsto \left(\pm\cos\frac{\pi|x|}{2}, \frac{\sin(\pi|x|/2)}{|x|}x\right), \qquad x \in D(V),$$

with the second factor interpreted smoothly at $x = 0$. They agree on the equator; x below denotes these disc coordinates. For a smooth map $\gamma : S(V) \to S^3$, define the principal right bundle P_γ by gluing $U_N \times S^3$ to $U_S \times S^3$ with the identification

$$(y, v_N)_N \sim (y, \gamma(y)v_N)_S.$$

The transition is extended independently of the signed radial variable on a collar of the equator; this specifies the smooth bundle structure. The right action is $(x, v_i)h = (x, v_i h)$ in either chart.

For a conjugation-equivariant transition $\gamma = \theta$, the local formula

$$q \star (x, v_i) = (\rho(q)x, qv_i)$$

defines a global free left action on P_θ. Indeed, $\theta(\rho(q)y)qv_N = q\theta(y)v_N$, which is the compatibility condition on the overlap. It commutes with the principal right action. The quotient coordinate is $\rho(v_i^{-1})x$. On the equator the change of quotient coordinates is therefore $J_{\theta^{-1}}(y) = \rho(\theta(y)^{-1})y$.

For an invariant transition $\gamma = b$, use instead

$$q \star (x, v_i) = (\rho(q)x, v_i q^{-1}). \tag{6.6}$$

Invariance of b makes this formula compatible with the transition. It is a free left action, since $v_i q^{-1} = v_i$ forces $q = 1$. Its quotient coordinate is $\rho(v_i)x$, and the change of quotient coordinates on the equator is $L_b(y) = \rho(b(y))y$. This action need not commute with the principal right action. The same formulas hold in polar coordinates $x = sy$ and extend to $x = 0$.

6.3. **Quotient metrics and boundary coordinates.** We compute the quotient metric on a fixed radial level.

Lemma 6.4 (The quotient metric)**.** *Let $f, r > 0$. On $S(V) \times S^3$ use the product metric $f^2 h_{S(V)} + r^2 h_{S^3}$ and either diagonal action described above. Use the section $y \mapsto (y, 1)$ to identify the quotient with $S(V)$. Write*

$$D_y = r^2 I + f^2 K_y^* K_y, \qquad C_y = I + \frac{f^2}{r^2} K_y K_y^*.$$

For $Y, Z \in T_y S(V)$ the quotient metric is

$$\begin{aligned} h_{f,r}(Y, Z) &= f^2 \left\langle C_y^{-1} Y, Z \right\rangle \\ &= f^2 \langle Y, Z \rangle - f^4 \left\langle D_y^{-1} K_y^* Y, K_y^* Z \right\rangle . \end{aligned} \tag{6.7}$$

In particular,

$$\frac{f^2 r^2}{r^2 + f^2 L_\rho^2} h_{S(V)} \;\le\; h_{f,r} \;\le\; f^2 h_{S(V)}. \tag{6.8}$$

The angular component of the horizontal lift of Y, measured in the scaled orthonormal frame, is $X = f C_y^{-1} Y$.

Proof. Fiber inversion identifies the two actions isometrically and fixes the section, so it suffices to use the diagonal action $q \star (x, v) = (\rho(q)x, qv)$. Project the section vector $(Y, 0)$ orthogonally away from the orbit $(K_y \xi, \xi)$. Orthogonality gives

$$\xi = f^2 D_y^{-1} K_y^* Y, \qquad (Y, 0)^{\mathrm{hor}} = (Y - K_y \xi, -\xi).$$

The identity

$$I - f^2 K_y D_y^{-1} K_y^* = C_y^{-1}$$

follows by multiplication by C_y. Taking the squared norm of this horizontal lift, and then polarizing, proves (6.7). Since $I \le C_y \le (1 + f^2 L_\rho^2 / r^2) I$, it also proves (6.8). Since $D_y \ge r^2 I$ and $C_y \ge I$, their inverses depend smoothly on y. □

Lemma 6.5 (The metric on a boundary collar)**.** *Let γ be either type of transition in the preceding construction, and let t be a collar coordinate. Suppose γ is independent of t and the northern-gauge metric is $dt^2 + f(t)^2 h_{S(V)} + r(t)^2 h_{S^3}$, where $f, r > 0$. In the southern gauge $v_S = \gamma(y) v_N$, the connection potential is $A_S = -d\gamma\, \gamma^{-1}$. Put $\phi = J_{\gamma^{-1}}$ in the conjugation-equivariant case and $\phi = L_\gamma$ in the invariant case. Then the quotient metric in the coordinates extending over the southern disc is*

$$\bar{G}_S = dt^2 + (\phi^{-1})^* h_{f(t), r(t)}. \tag{6.9}$$

Proof. The change of gauge $F(t, y, v_N) = (t, y, \gamma(y) v_N)$ satisfies

$$F^* \left(\mathrm{Ad}_{v_S^{-1}} A_S + v_S^{-1} dv_S \right) = v_N^{-1} dv_N.$$

Thus the change of trivialization is an isometry between the two expressions for the total-space metric. It is equivariant for the corresponding star action and, in quotient coordinates, becomes $(t, y) \mapsto (t, \phi(y))$. The quotient metric in lemma 6.4 therefore gives (6.9). Since ϕ is independent of t, radial differentiation commutes with this pullback. □

6.4. **Curvature of the northern quotient.** The northern metric is a doubly warped product. The following criterion controls its curvature on planes horizontal for ϖ.

Proposition 6.6 (Positive curvature of the northern quotient)**.** *Let $\ell > 0$ and consider*

$$G_N = ds^2 + F(s)^2 h_{S(V)} + r(s)^2 h_{S^3}, \qquad 0 \le s \le \ell, \tag{6.10}$$

where $F, r \in C^\infty([0,\ell])$, $F(0) = 0$, $F'(0) = 1$, and F has an odd smooth extension near zero. Assume $r > 0$ is constant near zero, $F' > 0$, and $0 \le r' < 1$. For a real number a, write $a_+ = \max\{a, 0\}$. On $s > 0$ let S^3 act freely by

$$q \star (y, u) = (\rho(q)y, qu). \tag{6.11}$$

Suppose the two functions

$$\begin{aligned} c_{\mathrm{rad}} &= -\frac{F''}{F} - \frac{L_\rho^2 F^2 (r'')_+}{r^3} > 0, \\ c_{\mathrm{ang}} &= \frac{1 - F'^2}{F^2} - \frac{2L_\rho^2 F F' r'}{r^3} > 0 \end{aligned} \tag{6.12}$$

extend continuously and positively to $s = 0$. Then G_N extends smoothly across the center and the quotient is a smooth positively curved disc. Set $F_a = F(\ell)$, $r_{\min} = \min r$, and $m_N = \min_{[0,\ell]} \min\{c_{\mathrm{rad}}, c_{\mathrm{ang}}\}$. Its sectional curvature is at least

$$\frac{m_N}{(1 + L_\rho^2 F_a^2 / r_{\min}^2)^2} > 0.$$

Proof. In Cartesian polar coordinates $x = sy$, the quotient is identified with $D(V)$ by $[(x, u)] \mapsto \rho(u^{-1})x$, after a fixed radial rescaling. The global section is $x \mapsto (x, 1)$. Right multiplication on the fiber commutes with (6.11), so it suffices to compute at fiber coordinate $u = 1$. Write angular and fiber components in orthonormal frames for the *scaled* metrics $F^2 h_{S(V)}$ and $r^2 h_{S^3}$. The orbit vector associated with ξ has components $(FK_y\xi, r\xi)$ in these frames. A vector horizontal for ϖ consequently has the form

$$x_0 \partial_s + X + T_y X, \qquad T_y = -\frac{F}{r} K_y^*, \qquad \|T_y\|_{\mathrm{op}} \le \frac{L_\rho F}{r}, \tag{6.13}$$

where L_ρ is fixed as in (6.5).

Writing the infinitesimal action as $\mathcal{O}_y \xi = (FK_y\xi, r\xi)$, the orthogonal projection onto the horizontal space is

$$\mathcal{O}_y^* \mathcal{O}_y = r^2 I + F^2 K_y^* K_y \ge r^2 I, \qquad \Pi_y = I - \mathcal{O}_y (\mathcal{O}_y^* \mathcal{O}_y)^{-1} \mathcal{O}_y^*. \tag{6.14}$$

The lower bound makes Π_y smooth. The radial direction is horizontal.

Fix $s > 0$ and $y \in S(V)$, and write $T = T_y$. Take horizontal vectors $P = x_0 \partial_s + X + TX$ and $Q = y_0 \partial_s + Y + TY$. Put $A = x_0 Y - y_0 X$ and $B = X \wedge Y$. The radial–angular, angular–angular, radial–fiber, angular–fiber and fiber–fiber components of $P \wedge Q$ are, respectively, A, B, TA, $X \otimes TY - Y \otimes TX$, and $TX \wedge TY$. The curvature operator is diagonal on these five blocks, so

$$\begin{aligned} \langle R^{G_N}(P, Q)Q, P \rangle_{G_N} = & -\frac{F''}{F}|A|^2 + \frac{1 - F'^2}{F^2}|B|^2 - \frac{r''}{r}|TA|^2 \\ & - \frac{F'r'}{Fr}|X \otimes TY - Y \otimes TX|^2 + \frac{1 - r'^2}{r^2}|TX \wedge TY|^2. \end{aligned} \tag{6.15}$$

The mixed component satisfies

$$|X \otimes TY - Y \otimes TX|^2 \le 2\,\|T\|_{\mathrm{op}}^2\,|X \wedge Y|^2.$$

If X, Y are dependent, both sides vanish. Otherwise replace them by an orthonormal basis e_1, e_2 of their span and divide by $|X\wedge Y|^2$. The left side becomes $|Te_1|^2+|Te_2|^2 \le 2\|T\|_{\mathrm{op}}^2$. Write $\mathcal{K}(P,Q) = \langle R^{G_N}(P,Q)Q, P\rangle_{G_N}$. The last term of (6.15) is nonnegative because $0 \le r' < 1$. The radial–fiber term is bounded below by $-(r'')_+\|T\|_{\mathrm{op}}^2|A|^2/r$. Since $F' > 0$ and $r' \ge 0$, the angular–fiber coefficient is nonpositive, so the preceding upper bound on its area gives a lower bound for its curvature contribution. Using $\|T\|_{\mathrm{op}} \le L_\rho F/r$ in these two estimates yields

$$\begin{aligned}\mathcal{K}(P,Q) \ge\ & \left(-\frac{F''}{F} - \frac{L_\rho^2 F^2 (r'')_+}{r^3}\right)|A|^2 \\ & + \left(\frac{1-F'^2}{F^2} - \frac{2L_\rho^2 F F' r'}{r^3}\right)|B|^2. \end{aligned} \tag{6.16}$$

The projected bivector is

$$(x_0\partial_s + X) \wedge (y_0\partial_s + Y) = \partial_s \wedge A + B.$$

If $A = B = 0$, the two projected vectors are dependent. Since the horizontal space is the graph of a linear map over this projection, P, Q would also be dependent. Thus every horizontal two-plane has a positive numerator under the stated hypotheses. At $s = 0$, the metric G_N is smooth because F has an odd smooth extension, $F'(0) = 1$, and r is constant nearby. In Cartesian disc coordinates, the orbit at the center is contained in the fiber, so the horizontal space is the base tangent space. The sectional curvature of G_N on such a plane is the common positive limit $-F'''(0)$ of the two base curvature coefficients.

With m_N and $r_{\min}$ as in the statement, the two coefficients in (6.16) are at least $m_N > 0$. The map $(x_0, X) \mapsto (x_0, X, TX)$ has norm at most $\sqrt{1 + \|T\|_{\mathrm{op}}^2}$. Taking its exterior square gives

$$|P \wedge Q|^2 \le (1 + L_\rho^2 F_a^2/r_{\min}^2)^2\big(|A|^2 + |B|^2\big). \tag{6.17}$$

Consequently, on every plane horizontal for ϖ, including at the center, the sectional curvature of G_N is at least

$$c_N = \frac{m_N}{(1 + L_\rho^2 F_a^2/r_{\min}^2)^2} > 0. \tag{6.18}$$

O'Neill's formula [14] gives the same lower bound on the northern quotient.

□

Appendix B constructs F and r with prescribed boundary radii and logarithmic derivative r'/r.

6.5. **Construction and gluing of the disc metrics.** For either transition function, choose the latitude a, cutoff χ and principal connection ω first. Their curvature bounds fix M_0, M_1 and then Λ, A_0. Next choose the cutoff ζ and δ, F, ℓ in Appendix B; the common fiber scale ε is chosen last, using both the connection bound and ℓ. All these choices may depend on the fixed transition and representation.

Theorem 6.7 (Invariant transition functions)**.** *Let $b : S(V) \to S^3$ be smooth and ρ-invariant. Let $P_b \to S(\mathbb{R} \oplus V)$ be the principal right S^3-bundle with equatorial transition $v_S = b(y)v_N$, equipped with the action (6.6). Its quotient admits a smooth Riemannian metric with strictly positive sectional curvature. With the disc parametrizations specified above, it is*

$$P_b/S^3_\star \cong D^n_N \cup_{L_b} D^n_S, \qquad n = \dim V. \tag{6.19}$$

Proof. The quotient and the attaching map. For the action of Section 6.2, the local quotient of $D(V) \times S^3$ is the smooth disc $D(V)$, with quotient map $(x, v) \mapsto \rho(v)x$ and section $x \mapsto (x, 1)$. In either polar trivialization define the quotient angular coordinate

$$y_i = \rho(v_i)y. \tag{6.20}$$

It is invariant under $\star$. On the overlap,

$$y_S = \rho(b(y)v_N)y = \rho(b(y_N))y_N = L_b(y_N),$$

where the middle equality uses the ρ-invariance of b. This proves (6.19).

The southern connection. Fix

$$\frac{\pi}{2} < a < b_0 < \pi.$$

Transport the equatorial transition radially to the latitude $t = a$. The two base pieces $0 \le t \le a$ and $a \le t \le \pi$ are still standard discs with the transported parametrizations. Radial reparameterization does not change their angular clutching map. Use the round southern base disc

$$g_B = \mathrm{d}t^2 + \sin^2 t\, h_{S(V)}, \qquad a \le t \le \pi. \tag{6.21}$$

Choose a smooth cutoff $\chi : [a, \pi] \to [0, 1]$ that is zero near $t = a$ and one for $t \ge b_0$. On the punctured cap $a \le t < \pi$, pull b back independently of t and put, in the northern and southern gauges,

$$A_N = \chi b^{-1}\,\mathrm{d}b, \qquad A_S = (\chi - 1)\,\mathrm{d}b\, b^{-1}. \tag{6.22}$$

Since $\chi = 1$ near π, extend A_S by zero across the south pole. The relation

$$A_N = b^{-1}A_S b + b^{-1}\,\mathrm{d}b$$

shows that these potentials define a smooth principal connection: in the ith trivialization its connection form is $\omega_i = \mathrm{Ad}_{v_i^{-1}} A_i + v_i^{-1}dv_i$. It is a product connection near the boundary in the northern gauge and near the center in the southern gauge. Put $a_b = b^{-1}db$, pulled back independently of t, and use the convention $(a_b \wedge a_b)(Y, Z) = [a_b(Y), a_b(Z)]$. The Maurer–Cartan identity $da_b + a_b \wedge a_b = 0$ gives the local curvature forms

$$\Omega_N = \chi'\, dt \wedge a_b + \chi(\chi - 1)a_b \wedge a_b, \qquad \Omega_S = \mathrm{Ad}_b\, \Omega_N. \tag{6.23}$$

The curvature vanishes where χ is identically 0 or 1. Thus Ω and $D\Omega$ are supported in the fixed annulus $[a, b_0] \times S(V)$, where $\sin t$ is bounded away from zero. Their norms, taken with respect to the fixed round base and Lie-algebra metrics, are bounded independently of r and ε.

Let $B_q(y, v) = (\rho(q)y, v)$ and let $R_h(y, v) = (y, vh)$ denote the principal right action. The invariance of b makes B_q compatible with the transition and gives

$B_q^* A_i = A_i$ in both gauges. Thus $B_q^*\omega = \omega$, whereas $R_h^*\omega = \mathrm{Ad}_{h^{-1}}\omega$. Since the $\star$-action is $B_q R_{q^{-1}}$, it follows that

$$(q\star)^*\omega = \mathrm{Ad}_q\,\omega, \qquad (q\star)^* Q_{\mathfrak{g}}(\omega,\omega) = Q_{\mathfrak{g}}(\omega,\omega). \tag{6.24}$$

Let M_0, M_1 be the finite curvature bounds of this fixed connection. Choose Λ as in theorem 6.3 and choose $A_0 > 0$ with

$$A_0\,|\cos a| \geq \Lambda.$$

The round height function satisfies $\mathrm{Hess}(\cos t) = -\cos t\, g_B$. Thus $\varphi(t) = -A_0\cos t$ satisfies

$$-\mathrm{Hess}\,\varphi = -A_0\cos t\, g_B \geq A_0|\cos a|\, g_B \geq \Lambda g_B, \qquad a \leq t \leq \pi.$$

Theorem 6.3 therefore makes

$$G_S = \mathrm{d}t^2 + \sin^2 t\, h_{S(V)} + r_S(t)^2 Q_{\mathfrak{g}}(\omega,\omega), \qquad r_S(t)^2 = \varepsilon e^{-\varepsilon A_0\cos t}, \tag{6.25}$$

a metric of strictly positive sectional curvature for every $0 < \varepsilon \leq \varepsilon_{\mathrm{conn}}$. The action is free by its definition; (6.24) and invariance of the round base and r_S show that it is isometric. O'Neill's formula gives positive sectional curvature on the southern quotient disc.

The northern metric. Write $r_S(t) = \sqrt{\varepsilon}\, e^{-\varepsilon A_0\cos t/2}$ and choose F, r, ℓ by Appendix B with boundary data $F_a = \sin a$, $r_a = r_S(a)$, and $q_s = r_S'(a)/r_a$. Fiber inversion $u = v^{-1}$ is an isometry of the round fiber and converts the action (6.6) to (6.11) while fixing the section $x \mapsto (x, 1)$. Proposition 6.6 therefore gives a smooth strictly positively curved quotient disc, including its center. These choices also give $F(\ell) = F_a$, $r(\ell) = r_a$, and $r'(\ell)/r_a = q_s$.

Boundary data and gluing. In the northern trivialization, both total-space boundary metrics are

$$F_a^2 h_{S(V)} + r_a^2 h_{S^3}, \qquad F_a = \sin a,$$

and the $\star$-action is the same. In the coordinates extending over the discs, the quotient boundary metrics therefore satisfy

$$h_N = L_b^* h_S. \tag{6.26}$$

On a southern collar, lemma 6.5 gives

$$\bar{G}_S = dt^2 + (L_b^{-1})^* h_{\sin t, r_S(t)}.$$

On the northern collar the quotient is $ds^2 + h_{F(s),r(s)}$. The angular identification L_b is independent of both radial coordinates. Put

$$\mu_N = \frac{F'(\ell)}{F_a} > 0, \qquad \mu_S = \cot a < 0, \qquad q_s = \frac{r'(\ell)}{r_a} = \frac{r_S'(a)}{r_a}.$$

In the common boundary coordinates, write $h = F^2 C^{-1}$, where $C = I + (F^2/r^2) K_y K_y^*$. For $\mu = F'/F$ and $q = r'/r$, $C' = 2(\mu - q)(C - I)$ commutes with C. Hence

$$\tfrac{1}{2} h^{-1} h' = qI + (\mu - q) C^{-1}. \tag{6.27}$$

The outward normals are $+\partial_s$ on the northern disc and $-\partial_t$ on the southern disc. Their shape endomorphisms are $q_s I + (\mu_N - q_s) C_y^{-1}$ and $-q_s I - (\mu_S - q_s) C_y^{-1}$, respectively, where $C_y = I + (F_a^2/r_a^2) K_y K_y^*$. Their sum is

$$h_N^{-1}\left(B_N + L_b^* B_S\right) = (\mu_N - \mu_S) C_y^{-1}. \tag{6.28}$$

Since $C_y \le (1 + L_\rho^2 F_a^2/r_a^2)I$, we obtain

$$B_N + L_b^* B_S \;\ge\; \frac{\mu_N - \mu_S}{1 + L_\rho^2 F_a^2/r_a^2}\, h_N > 0. \tag{6.29}$$

Together with (6.26), this verifies the hypotheses of theorem 6.1 and proves the result. □

Theorem 6.8 (Conjugation-equivariant transition functions)**.** *Let $\rho : S^3 \to O(V)$ be an orthogonal representation with $\dim V = n \ge 3$, and let $\theta : S(V) \to S^3$ be smooth and conjugation equivariant. Form $P_\theta \to S(\mathbb{R} \oplus V)$ with transition $v_S = \theta(y)v_N$ and the action $q \star (x, v) = (\rho(q)x, qv)$. Then the quotient*

$$P_\theta / S^3_\star \cong D^n_N \cup_{J_{\theta^{-1}}} D^n_S$$

admits a smooth metric of strictly positive sectional curvature.

Proof. This construction also appears in [5, Proposition 5.1] and [6, Theorem 4.1 and Remark 4.4]. Conjugation equivariance makes $v_S = \theta(y)v_N$ compatible with the action $q \star (x, v) = (\rho(q)x, qv)$. Its quotient disc coordinate is $y_i = \rho(v_i^{-1})y$, including the analogous Cartesian coordinate at the center. Conjugation equivariance gives

$$y_S = \rho(\theta(y_N)^{-1})y_N = J_{\theta^{-1}}(y_N).$$

Choose $\pi/2 < a < b_0 < \pi$ and a smooth cutoff $\chi : [a, \pi] \to [0, 1]$ that is zero near a and one for $t \ge b_0$. On the punctured cap take $A_N = \chi\theta^{-1}d\theta$ and $A_S = (\chi - 1)d\theta\,\theta^{-1}$, extending A_S by zero over the pole. Under the base action their pullbacks are $\mathrm{Ad}_q A_i$; simultaneous left multiplication of v_i by q then cancels this adjoint factor in $\omega_i = \mathrm{Ad}_{v_i^{-1}} A_i + v_i^{-1}dv_i$. Hence the principal connection, and therefore its connection metric, is invariant under $\star$. Formula (6.23) applies with b replaced by θ, so its curvature bounds are again fixed before the fiber scale.

Let $M_0 = \sup|\Omega|$ and $M_1 = \sup|D\Omega|$ for this connection. Choose Λ by (6.1), and then choose A_0 with $A_0|\cos a| \ge \Lambda$. The southern connection metric is positive by theorem 6.3. With these data fixed, choose δ, F, ℓ and finally the common scale ε as in Appendix B. The northern action is already (6.11), so proposition 6.6 gives the northern metric. The endpoint conditions make both total-space boundary metrics equal in the northern trivialization. Thus $h_N = J_{\theta^{-1}}^* h_S$ on the boundary. The southern collar metric is $dt^2 + (J_{\theta^{-1}}^{-1})^* h_{\sin t, r_S(t)}$ by lemma 6.5. Applying (6.27) with the two outward normals gives

$$h_N^{-1}(B_N + J_{\theta^{-1}}^* B_S) = (\mu_N - \mu_S)C_y^{-1} > 0.$$

Theorem 6.1 now gives a smooth positively curved metric on the stated clutched manifold. □

Remark 6.9 (The induced group action)*.* For a conjugation-equivariant θ, the principal right action commutes with $\star$ and induces a left action on the quotient by $h \cdot [p] = [ph^{-1}]$. In each disc coordinate this is $y \mapsto \rho(h)y$; equivalently, $J_{\theta^{-1}}$ is ρ-equivariant. The source metrics are principal-right-invariant, so both quotient disc metrics are invariant under this action. Corollary 6.2 gives an invariant metric of positive sectional curvature. If ρ is nontrivial, its effective image is S^3 or $SO(3)$. The Lawson–Yau theorem applied to this nonabelian group action gives a metric of positive scalar curvature [13]. Hence a spin quotient has vanishing Hitchin invariant. If ρ is trivial, the attaching map is the identity and the quotient is the standard sphere.

7. Completion of the proof

We apply the preceding metric constructions to the three attaching maps. The required representation bounds are as follows.

7.1. **The two representations.** For $V_\kappa = \mathbb{R} \oplus \mathbb{H}^2$ and $y = (t, w, z)$, differentiation gives

$$K_y \xi = (0, \xi w, \xi z). \tag{7.1}$$

Multiplicativity of the quaternionic norm yields

$$|K_y \xi|^2 = (|w|^2 + |z|^2)|\xi|^2 = (1 - t^2)|\xi|^2, \tag{7.2}$$

and hence

$$K_y^* K_y = (1 - t^2) I, \qquad \|K_y\|_{\mathrm{op}} \leq 1. \tag{7.3}$$

For $V_\lambda = \mathbb{R}^2 \oplus \mathbb{H} \oplus \operatorname{Im} \mathbb{H}$ and $y = (a, w, z)$,

$$K_y \xi = (0, \xi w, [\xi, z]). \tag{7.4}$$

Since $[\xi, z] = 2\xi \times z$, one obtains

$$K_y^* K_y = |w|^2 I + 4\Big(|z|^2 I - z \otimes z\Big). \tag{7.5}$$

If $z \neq 0$, the decomposition $\operatorname{Im} \mathbb{H} = \mathbb{R} z \oplus z^\perp$ gives the eigenvalue $|w|^2$ on $\mathbb{R}z$ and $|w|^2 + 4|z|^2$ on the two-dimensional space $z^\perp$. If $z = 0$, the Gram matrix is $|w|^2 I$, so its only eigenvalue has multiplicity three. In both cases, $\|K_y\|_{\mathrm{op}}^2 = |w|^2 + 4|z|^2 \leq 4$, since $|a|^2 + |w|^2 + |z|^2 = 1$. Thus

$$\|K_y\|_{\mathrm{op}} \leq 2. \tag{7.6}$$

The ranks are three for $w \neq 0$, two for $w = 0$, $z \neq 0$, and zero for $w = z = 0$.

7.2. **Application to the three sphere classes.** Table 2 lists the transition functions and the attaching maps in the specified disc coordinates. Its column headed "transition" records γ in $v_S = \gamma(y) v_N$. The first two transitions are conjugation equivariant and use the action $q \star (x, v) = (\rho(q)x, qv)$. The last is invariant and uses the action (6.6). The quotient-coordinate convention accounts for the inverses in the first two transitions.

sphere class	representation	transition	attaching map
κ	V_κ	β_κ^{-1}	J_{β_κ}
λ	V_λ	β_λ^{-1}	J_{β_λ}
$\kappa + \lambda$	V_κ	b_κ	L_{b_κ}

Table 2. Transition functions and attaching maps for the three non-standard classes.

Proof of theorem 1.1. The standard sphere has its round metric. Apply theorem 6.8 to the first two rows of Table 2; their smooth types are identified in propositions 4.3 and 4.8. Apply theorem 6.7 to the last row, whose type is given by corollary 5.4. Thus every class of $\ker \alpha$ admits positive sectional curvature.

Positive sectional curvature implies nonnegative sectional curvature and positive scalar curvature. Conversely, Hitchin's theorem gives $\alpha = 0$ from positive scalar

curvature, while proposition 2.1 gives $\alpha = 0$ from nonnegative sectional curvature. These implications prove all four equivalences. □

Remark 7.1. On the Kervaire sphere, the metric constructed here cannot be invariant under the full natural $SO(2) \times SO(5)$ action on its Brieskorn model. Grove–Verdiani–Wilking–Ziller exclude nonnegative sectional curvature for metrics invariant under that action, not for arbitrary metrics on the underlying manifold [7].

Appendix A. A quantitative connection-metric criterion

We prove theorem 6.3 using the curvature formulas and decomposable-bivector estimates of [9, Section 3]. Throughout this appendix, write $G = G_\varepsilon$ and $r = r_\varepsilon$. Horizontal vectors are identified with their projections when inserted into base tensors. The dual metric g_B^{-1} pairs base covectors.

A.1. **Frames and curvature blocks.** Write the base dimension as d and let X_i be the horizontal lifts, with respect to ω, of a base orthonormal frame normal at the point under consideration. Use the fundamental vector fields $\xi^\#$ of the principal right action defined in the bundle conventions. Choose a positively oriented orthonormal basis e_1, e_2, e_3 of $\operatorname{Im}\mathbb{H}$ and set $E_a = r^{-1} e_a^\#$, so that E_1, E_2, E_3 are unit vertical vectors. Write

$$[e_a, e_b] = c_{ab}{}^c e_c, \qquad c_{ab}{}^c = 2\epsilon_{abc}, \qquad \vartheta = d\log r, \qquad \mathsf{H} = -r^{-1}\operatorname{Hess} r.$$

Here ϵ_{abc} is the alternating symbol, with $\epsilon_{123} = 1$; $\vartheta^\sharp$ denotes the g_B-dual vector field of ϑ. Superscripts H, V denote the orthogonal projections to $\ker\omega$ and $\ker d\pi$. Write $\Omega_{ij} = \sum_b \Omega_{ij}^b e_b$ in the fixed Lie-algebra basis. Principal equivariance gives the vertical derivatives of these coefficient functions:

$$E_a(\Omega_{ij}^b) = -r^{-1} \sum_c c_{ac}{}^b \Omega_{ij}^c. \tag{A.1}$$

The bracket and Koszul identities are

$$\begin{aligned}
[X_i, X_j]^{\mathrm{V}} &= -r\Omega_{ij}^a E_a, & [X_i, E_a] &= -\vartheta_i E_a,\\
[E_a, E_b] &= r^{-1} c_{ab}{}^c E_c, & \nabla_{X_i} X_j &= (\nabla^B_i X_j)^{\mathrm{H}} - \tfrac{r}{2}\Omega_{ij}^a E_a,\\
\nabla_{X_i} E_a &= \tfrac{r}{2}\Omega_{ij}^a X_j, & \nabla_{E_a} X_i &= \tfrac{r}{2}\Omega_{ij}^a X_j + \vartheta_i E_a,\\
\nabla_{E_a} E_b &= \tfrac{1}{2r} c_{ab}{}^c E_c - \delta_{ab}\vartheta_i X_i.
\end{aligned} \tag{A.2}$$

Repeated indices are summed over their full ranges. Put $R_{ABCD} = \langle R(A,B)C, D\rangle_G$. Differentiating (A.2), using (A.1), gives

$$\begin{aligned}
R_{iabj} &= \mathsf{H}_{ij}\delta_{ab} + \frac{r^2}{4}\sum_l \Omega_{il}^b \Omega_{jl}^a - \frac{1}{4}\sum_c c_{ab}{}^c \Omega_{ij}^c,\\
R_{ijab} &= \frac{r^2}{4}\sum_l (\Omega_{il}^a\Omega_{jl}^b - \Omega_{jl}^a\Omega_{il}^b) + \frac{1}{2}\sum_c c_{ab}{}^c\Omega_{ij}^c,\\
R_{ijka} &= -\frac{r}{2}\Big((D_i\Omega)_{jk}^a - (D_j\Omega)_{ik}^a\Big) - \frac{r}{2}(\vartheta_i\Omega_{jk}^a - \vartheta_j\Omega_{ik}^a) + r\vartheta_k\Omega_{ij}^a,\\
R_{iabc} &= \frac{r}{2}\sum_l \vartheta_l(\delta_{ab}\Omega_{il}^c - \delta_{ac}\Omega_{il}^b),\\
R_{abcd} &= (r^{-2} - |\vartheta|^2)(\delta_{bc}\delta_{ad} - \delta_{ac}\delta_{bd}).
\end{aligned} \tag{A.3}$$

In R_{iabj}, the vertical derivative and the vertical Levi–Civita term contribute respectively $-1/2$ and $+1/4$ times $\sum_c c_{ab}{}^c \Omega^c_{ij}$, giving $-1/4$. The purely horizontal block is

$$R_{ijkl} = R^B_{ijkl} + \frac{r^2}{4}\Big(-\langle \Omega_{jk}, \Omega_{il}\rangle + \langle \Omega_{ik}, \Omega_{jl}\rangle + 2\langle \Omega_{ij}, \Omega_{kl}\rangle\Big). \tag{A.4}$$

In particular, for horizontal X, Y,

$$\langle R(X,Y)Y, X\rangle_G = \langle R^B(X,Y)Y, X\rangle_{g_B} - \tfrac{3}{4} r^2 |\Omega(X,Y)|^2. \tag{A.5}$$

The remaining components follow from the curvature symmetries.

A.2. **A lower bound on every decomposable bivector.** Let $P = X + U$ and $Q = Y + V$, where X, Y are horizontal for ω and U, V are components in the unit vertical frame. In expressions such as $[U, V]$, identify $\sum_a U_a E_a$ with $\sum_a U_a e_a \in \operatorname{Im}\mathbb{H}$ and use the bracket of this fixed Lie algebra. Decompose

$$P \wedge Q = H + B + W, \qquad H = X \wedge Y, \quad B = X \otimes V - Y \otimes U, \quad W = U \wedge V,$$

and put $h = |H|$, $m = |B|$, $k = |W|$. These are orthogonal components, so $|P \wedge Q|^2 = h^2 + m^2 + k^2$. We also regard the mixed tensor B as the linear map from the vertical coefficient space to the horizontal space given by

$$B\xi = X\langle V, \xi\rangle - Y\langle U, \xi\rangle.$$

Thus $B^* Z = V\langle X, Z\rangle_G - U\langle Y, Z\rangle_G$ for horizontal Z, and $\|B\|_{\mathrm{HS}} = m$, where the Hilbert–Schmidt norm is the square root of the sum of squared matrix entries in orthonormal bases. Use the curvature bilinear form $\mathcal{R}(A\wedge B, C\wedge D) = \langle R(A,B)D, C\rangle_G$ and put $\mathcal{K}(P \wedge Q) = \langle R(P,Q)Q, P\rangle_G$.

The terms linear in Ω combine as follows. The mixed block and horizontal–vertical block in (A.3) give, respectively, $-\frac{1}{2}\langle \Omega(X,Y), [U,V]\rangle$ and $-\langle \Omega(X,Y), [U,V]\rangle$. Their sum is

$$-\frac{3}{2}\langle \Omega(X,Y), [U,V]\rangle. \tag{A.6}$$

For $B_{ia} = X_i V_a - Y_i U_a$, the mixed contribution follows from

$$\sum_{i,j,a,b,c} B_{ia} B_{jb} c_{ab}{}^c \Omega^c_{ij} = 2\langle \Omega(X,Y), [U,V]\rangle.$$

Since $|[U,V]| = 2|U \wedge V|$, the absolute value of (A.6) is at most $3M_0 hk$.

To combine the quadratic terms, for a vertical coefficient vector U define the base covector

$$\Omega_U(X)(Z) = \langle \Omega(X,Z), U\rangle.$$

The entire part of the sectional numerator that is quadratic in Ω is

$$\begin{aligned} \mathcal{Q}_{\Omega^2} = & -\frac{3r^2}{4}|\Omega(X,Y)|^2 + \frac{r^2}{4}|\Omega_V(X) + \Omega_U(Y)|^2 \\ & - r^2 \langle \Omega_U(X), \Omega_V(Y)\rangle_{g_B^{-1}}. \end{aligned} \tag{A.7}$$

Indeed, the quadratic mixed block contributes

$$\frac{r^2}{4}\Big(|\Omega_V(X)|^2 + |\Omega_U(Y)|^2 - 2\langle \Omega_U(X), \Omega_V(Y)\rangle_{g_B^{-1}}\Big),$$

and the horizontal–vertical cross block contributes

$$\frac{r^2}{2}\Big(\langle \Omega_V(X), \Omega_U(Y)\rangle_{g_B^{-1}} - \langle \Omega_U(X), \Omega_V(Y)\rangle_{g_B^{-1}}\Big).$$

Their sum, together with (A.5), proves (A.7).

For estimates, choose an orthonormal basis P, Q of the two-plane that diagonalizes the symmetric form $(Z, Z') \mapsto \langle \mathrm{pr_H}\, Z, \mathrm{pr_H}\, Z' \rangle_G$ restricted to that plane, where $\mathrm{pr_H}$ is the horizontal orthogonal projection. Then

$$\langle X, Y \rangle_G = \langle U, V \rangle = 0, \qquad h = |X||Y|, \quad k = |U||V|, \quad m^2 = |X|^2|V|^2 + |Y|^2|U|^2. \tag{A.8}$$

Under a change of spanning basis, the sectional numerator and each area squared scale by the square of the determinant. The resulting estimates therefore hold for any independent spanning pair. Since $|\Omega_U(X)| \le M_0|X||U|$, discarding the square in (A.7) gives

$$\mathcal{Q}_{\Omega^2} \ge -\tfrac{3}{4} r^2 M_0^2 h^2 - r^2 M_0^2 hk. \tag{A.9}$$

To compute the mixed terms with three horizontal arguments, evaluate the R_{ijka} block in (A.3) at its repeated horizontal argument. For $Z = X$ or $Z = Y$ and any vertical vector U_0, alternation of Ω and $D\Omega$ in their two-form arguments gives

$$\langle R(X, Y)Z, U_0 \rangle_G = \frac{r}{2} \langle (D_Z \Omega)(X, Y), U_0 \rangle + \frac{3r}{2} \vartheta(Z) \langle \Omega(X, Y), U_0 \rangle.$$

The cross contribution to the curvature quadratic form on $H+B$ is $2\langle R(X, Y)Y, U \rangle_G - 2\langle R(X, Y)X, V \rangle_G$. Consequently,

$$\begin{aligned} \mathcal{Q}_{3\mathrm{H}} = r\Big(&\langle (D_Y \Omega)(X, Y), U \rangle - \langle (D_X \Omega)(X, Y), V \rangle \Big) \\ &+ 3r\Big(\vartheta(Y) \langle \Omega(X, Y), U \rangle - \vartheta(X) \langle \Omega(X, Y), V \rangle \Big). \end{aligned} \tag{A.10}$$

For the first line, contract the array $((D_i \Omega)(X, Y)^a)_{ia}$ with $-B_{ia}$; its norm is at most $M_1 h$. For the second, contract $\Omega(X, Y)$ with $\vartheta(Y)U - \vartheta(X)V = -B^* \vartheta^\sharp$. Consequently

$$|\mathcal{Q}_{3\mathrm{H}}| \le r(M_1 + 3|\vartheta| M_0) hm.$$

For the mixed terms with one horizontal argument, use W also for the skew endomorphism of the vertical coefficient space associated with $U \wedge V$:

$$W\xi = U\langle V, \xi \rangle - V\langle U, \xi \rangle.$$

In orthonormal bases put

$$W_{ab} = U_a V_b - V_a U_b, \qquad Z_{ia} = \sum_l \vartheta_l \Omega^a_{il}.$$

The matrices B, W, Z have sizes $d \times 3$, 3×3, and $d \times 3$, respectively. Write $\langle A, B \rangle_{\mathrm{HS}} = \mathrm{tr}(A^* B)$ for the Hilbert–Schmidt inner product. The terms with one horizontal argument are

$$2\langle R(X, V)V, U \rangle_G - 2\langle R(Y, U)V, U \rangle_G.$$

Substitution of the R_{iabc} block in (A.3) turns this expression into

$$r \sum_{i,a} Z_{ia} \Big(|V|^2 X_i U_a + |U|^2 Y_i V_a - \langle U, V \rangle (X_i V_a + Y_i U_a) \Big).$$

The coefficient in parentheses is $-(BW)_{ia}$, so

$$\mathcal{Q}_{1\mathrm{H}} = -r\langle BW, Z \rangle_{\mathrm{HS}}. \tag{A.11}$$

If U, V are dependent, $W = 0$. Otherwise W vanishes on their orthogonal complement and acts on their span as a rotation by a right angle multiplied by k. Hence $\|W\|_{\mathrm{op}} = k$ in both cases, and $\|BW\|_{\mathrm{HS}} \le \|B\|_{\mathrm{HS}} \|W\|_{\mathrm{op}} = mk$. If $\vartheta = 0$, then $Z = 0$. Otherwise

choose the first base frame vector to be $\vartheta^\sharp/|\vartheta|$. In this frame $Z_{ia} = |\vartheta|\Omega^a_{i1}$, so the stated $i<j$ norm gives

$$\|Z\|^2_{\mathrm{HS}} = |\vartheta|^2 \sum_{i=2}^{d}\sum_{a=1}^{3}(\Omega^a_{i1})^2 \le |\vartheta|^2 \sum_{i<j,a}(\Omega^a_{ij})^2 \le |\vartheta|^2 M_0^2.$$

Cauchy–Schwarz now gives $|\mathcal{Q}_{1\mathrm{H}}| \le r|\vartheta|M_0 km$. The Hessian term is $\sum_{i,j,a}\mathsf{H}_{ij}B_{ia}B_{ja}$. Here $\lambda_{\min}(\mathsf{H})$ denotes the least eigenvalue of the self-adjoint endomorphism $g_B^{-1}\mathsf{H}$, so this term is at least $\lambda_{\min}(\mathsf{H})m^2$. The vertical block is $(r^{-2}-|\vartheta|^2)k^2$.

Combining these estimates with (A.6) and the base sectional lower bound gives the estimate

$$\begin{aligned}\mathcal{K}(P\wedge Q) \ge{}& (\kappa_0 - \tfrac{3}{4}r^2M_0^2)h^2 + \lambda_{\min}(\mathsf{H})m^2 + (r^{-2}-|\vartheta|^2)k^2\\ &- r(M_1+3|\vartheta|M_0)hm - (3M_0+r^2M_0^2)hk - r|\vartheta|M_0km.\end{aligned} \tag{A.12}$$

For $C_0 = 16$, this implies the bound

$$\begin{aligned}\mathcal{K}(P\wedge Q) \ge{}& (\kappa_0 - C_0r^2M_0^2)h^2 + (r^{-2}-|\vartheta|^2)k^2\\ &+ (\lambda_{\min}(\mathsf{H}) - C_0r^2M_0^2)m^2\\ &- C_0r(M_1+|\vartheta|M_0)hm - C_0(M_0+r^2M_0^2)hk\\ &- C_0r|\vartheta|M_0km.\end{aligned} \tag{A.13}$$

Since $H = X\wedge Y$, the base sectional lower bound gives $\langle R^B(X,Y)Y,X\rangle_{g_B} \ge \kappa_0 h^2$.

A.3. **Estimates for the mixed terms.** Let $C_0 = 16$, let Λ satisfy (6.1), and assume $-\operatorname{Hess}\varphi \ge \Lambda g_B$. Define

$$P_* = \max\{1, \|\varphi\|_\infty\}, \qquad D_* = \max\{1, \|d\varphi\|_\infty\}, \qquad M_* = \max\{1, M_0\}.$$

An explicit sufficient scale is

$$\varepsilon_{\mathrm{conn}} = \frac{1}{2}\min\left\{1,\ \frac{\log 2}{P_*},\ \frac{1}{2M_*},\ \frac{1}{D_*M_*},\ \frac{\Lambda}{D_*^2},\ \frac{\kappa_0}{128C_0^2M_*^2},\ \left(\frac{1}{2C_0^2D_*^2M_*^2}\right)^{1/3}\right\}. \tag{A.14}$$

All denominators are positive, including when $M_0 = 0$.

For $0<\varepsilon\le\varepsilon_{\mathrm{conn}}$ set $r^2 = \varepsilon e^{\varepsilon\varphi}$. Then

$$\tfrac{1}{2}\varepsilon \le r^2 \le 2\varepsilon, \qquad \vartheta = \tfrac{1}{2}\varepsilon\, d\varphi, \qquad \mathsf{H} = -\tfrac{1}{2}\varepsilon\operatorname{Hess}\varphi - \tfrac{1}{4}\varepsilon^2 d\varphi\otimes d\varphi.$$

In particular, $\lambda_{\min}(\mathsf{H}) \ge \varepsilon\Lambda/4$. The threshold also gives

$$\begin{gathered} r^2M_0 \le 1, \qquad |\vartheta|M_0 \le 1, \qquad C_0r^2M_0^2 \le \kappa_0/4,\\ r^2|\vartheta|^2 \le 1/4, \qquad 64C_0^2r^2M_0^2 \le \kappa_0, \qquad 2C_0^2\varepsilon^3\|d\varphi\|_\infty^2M_0^2 \le 1.\end{gathered} \tag{A.15}$$

Young's inequality gives the following bounds for the three mixed terms in (A.13):

$$\begin{aligned} C_0r(M_1+|\vartheta|M_0)hm &\le \frac{\kappa_0}{8}h^2 + 4C_0^2\kappa_0^{-1}\varepsilon(M_1+1)^2m^2,\\ C_0(M_0+r^2M_0^2)hk &\le \frac{\kappa_0}{8}h^2 + \frac{8C_0^2M_0^2}{\kappa_0}k^2 \le \frac{\kappa_0}{8}h^2 + \frac{k^2}{8r^2},\\ C_0r|\vartheta|M_0km &\le \frac{k^2}{8r^2} + 2C_0^2r^4|\vartheta|^2M_0^2m^2 \le \frac{k^2}{8r^2} + \varepsilon m^2.\end{aligned} \tag{A.16}$$

The first and last estimates use $r^2 \le 2\varepsilon$ and $|\vartheta| \le \varepsilon\|d\varphi\|_\infty/2$. The horizontal and vertical coefficients left after these absorptions are at least $\kappa_0/2$ and $1/(2r^2)$. The mixed coefficient is at least

$$\varepsilon\left(\frac{\Lambda}{4} - 2C_0M_0^2 - 4C_0^2\kappa_0^{-1}(M_1+1)^2 - 1\right) \ge \frac{\varepsilon\Lambda}{8}$$

by (6.1). Thus

$$\boxed{\mathcal{K}(P\wedge Q) \ge \frac{\kappa_0}{2}h^2 + \frac{\varepsilon\Lambda}{8}m^2 + \frac{1}{2r^2}k^2.} \tag{A.17}$$

Since $r^2 \le 2\varepsilon$, division by $h^2+m^2+k^2$ gives (6.3), including at the boundary. This proves theorem 6.3.

Appendix B. Choice of the warping functions

We use the polynomial radial reparameterization of [5, Section 5.3]. The estimates below are adapted to our cutoff interval and to the boundary data in theorem 6.7. Fix $a \in (\pi/2, \pi)$, put $F_a = \sin a$, and let $L_* = \max\{1, L_\rho\}$, where $L_\rho = \sup_y \|K_y\|_{\mathrm{op}}$. After the southern connection and $A_0 > 0$ have been fixed, choose a smooth nondecreasing cutoff $\zeta : [0,\infty) \to [0,1]$ such that

$$\zeta = 0 \text{ on } [0,1], \qquad \zeta = 1 \text{ on } [2,\infty).$$

Write $M_\zeta = \sup|\zeta'| \ge 1$. Choose $\delta > 0$ with

$$\delta < F_a/4 \tag{B.1}$$

and

$$16L_*^2A_0F_a^2\delta^2 \le 1, \qquad 2000L_*^2A_0F_aM_\zeta\delta^3 \le 1. \tag{B.2}$$

For example, take

$$\delta = \frac{1}{2}\min\left\{\frac{F_a}{4},\ (16L_*^2A_0F_a^2)^{-1/2}, (2000L_*^2A_0F_aM_\zeta)^{-1/3}\right\}. \tag{B.3}$$

Define F by the strictly increasing polynomial relation

$$s = F + \frac{F^3}{3\delta^2}, \qquad F(0) = 0, \qquad F' = \left(1 + \frac{F^2}{\delta^2}\right)^{-1}, \tag{B.4}$$

and set

$$\ell = F_a + \frac{F_a^3}{3\delta^2}. \tag{B.5}$$

Then $F(\ell) = F_a$, $F' > 0$, $F \le s$, and $\ell > F_a > 4\delta$. The inverse of the odd polynomial in (B.4) is odd and smooth, since its derivative is everywhere positive. Writing $x = F/\delta$, direct differentiation gives

$$k_R := -\frac{F''}{F} = \frac{2}{\delta^2(1+x^2)^3}, \qquad k_A := \frac{1-F'^2}{F^2} = \frac{2+x^2}{\delta^2(1+x^2)^2}. \tag{B.6}$$

These formulas include their smooth positive limits at $F = 0$.

Use the same southern boundary radius and logarithmic derivative:

$$r_a^2 = \varepsilon e^{\varepsilon A_0|\cos a|}, \qquad q_s = \tfrac{1}{2}\varepsilon A_0F_a, \qquad d = r_aq_s.$$

Define

$$r(s) = r_a - d\int_s^\ell \zeta(v/\delta)\,\mathrm{d}v. \tag{B.7}$$

After δ, F, ℓ have been fixed, let $\varepsilon_S = \varepsilon_{\text{conn}}$ for the fixed southern data and choose

$$\varepsilon_* = \frac{1}{2}\min\left\{\varepsilon_S, \frac{\log 2}{A_0|\cos a|}, \frac{1}{A_0 F_a \ell}, \left(\frac{1}{2A_0F_a}\right)^{2/3}\right\}, \qquad 0 < \varepsilon \le \varepsilon_*. \tag{B.8}$$

The southern metric remains positive. Also $r_a^2 \le 2\varepsilon$, $q_s\ell \le 1/4$, and $d \le A_0F_a\varepsilon^{3/2}/\sqrt{2} < 1$. Consequently

$$\tfrac{3}{4}r_a \le r \le r_a, \qquad r' = d\zeta(s/\delta) \in [0,1), \qquad r'' = \frac{d}{\delta}\zeta'(s/\delta) \ge 0.$$

In particular r is constant near zero, and

$$\frac{d}{r^3} \le \frac{8q_s}{r_a^2} = 4A_0F_a e^{-\varepsilon A_0|\cos a|} \le 4A_0F_a. \tag{B.9}$$

For the angular condition in (6.12), the ratio of the term to be subtracted to k_A, for $F > 0$, satisfies

$$\frac{2L_\rho^2 FF'r'/r^3}{k_A} \le 8L_\rho^2 A_0F_a\delta^2 F\frac{1+x^2}{2+x^2} \le 8L_\rho^2A_0F_a^2\delta^2 \le \tfrac{1}{2}. \tag{B.10}$$

Thus $c_{\text{ang}} \ge k_A/2 > 0$. At the center the assertion follows by continuity, with $r' = 0$ nearby.

For the radial condition, r'' is supported in $[\delta, 2\delta]$. There $F \le 2\delta$ and hence

$$k_R \ge \frac{2}{125\delta^2}, \qquad \frac{L_\rho^2F^2r''}{r^3} \le 16L_\rho^2A_0F_aM_\zeta\delta \le \frac{1}{125\delta^2} \le \frac{k_R}{2}.$$

Outside this interval $r'' = 0$. Therefore $c_{\text{rad}} \ge k_R/2 > 0$ on the whole disc. Since $0 \le F \le F_a$ and $(2+x^2)(1+x^2) \ge 2$, the two estimates yield the explicit bound

$$m_N \ge m_N^{\text{poly}} := \frac{1}{\delta^2(1+F_a^2/\delta^2)^3} > 0. \tag{B.11}$$

Together with proposition 6.6 and $r_{\min} \ge r_a/2$ this implies

$$\sec_{\bar{G}_N} \ge \frac{1}{\delta^2(1+F_a^2/\delta^2)^3(1+4L_\rho^2F_a^2/r_a^2)^2} > 0.$$

Finally,

$$F(s) = s - \frac{s^3}{3\delta^2} + O(s^5), \qquad \lim_{s\to0}k_R = \lim_{s\to0}k_A = 2\delta^{-2}.$$

Since r is constant near zero, G_N is smooth and has positive curvature on horizontal planes at the center. O'Neill's formula then gives positive curvature of the quotient there. Since $\ell > 4\delta$, the cutoff equals one on a neighborhood of the northern boundary. Hence

$$F(\ell) = F_a, \qquad r(\ell) = r_a, \qquad r'(\ell)/r_a = q_s.$$

Appendix C. The Gromoll construction and stabilization

We use the model of framed homeomorphisms in [16, Sections 8–9] to compare the Gromoll construction with stabilization.

C.1. **Families of framed homeomorphisms.** Let $\mathscr{H}_m = \mathrm{Homeo}_\partial(D^m)$ be the topological group of homeomorphisms equal to the identity near the boundary, with the compact-open topology. We use the $GL_m(\mathbb{R})$ version $\mathscr{F}_m$ of the framed-homeomorphism model in [16, Sections 3.4 and 8]. Its local data are triples (h, A, H) with $h \in \mathscr{H}_m$, a continuous map $A : D^m \to GL_m(\mathbb{R})$, and a continuous family $H_{t,z}$ of invertible germs fixing zero in $\mathbb{R}^m$. Two representatives of a germ are identified when they agree on a neighborhood of zero; the family is understood as a germ along the zero section, so representatives and their inverses vary continuously in their parameters. The germ at z is a map from the normal coordinate over z to that over $h(z)$. The required endpoints are

$$H(0) = A, \qquad H(1) = \Delta h, \qquad (\Delta h)_z(v) = h(z+v) - h(z). \tag{C.1}$$

The inclusion $O_m \hookrightarrow GL_m(\mathbb{R})$ is a homotopy equivalence and induces the comparison with the O_m-framed model. For triples over h and k, composition has linear part $A_{k(z)}B_z$ and germ path $H_{t,k(z)} \circ K_{t,z}$, over $h \circ k$. All data are the identity on a boundary collar. The construction may be made in the simplicial germ model, so the statements below apply to compact parameter families and their homotopies.

Let $a \in D^r$ be a family parameter. A relative family (h_a, A_a, H_a) is extended by the identity in both variables. Its underlying product homeomorphism is

$$\widehat{h}(a, z) = (a, h_a(z)).$$

We associate to $\widehat{h}$ a triple in $\mathscr{F}_{m+r}$ with initial framing $I_r \oplus A_{a,z}$ as follows. First hold the parameter a fixed and use the path

$$(u, v) \longmapsto \Big(u, (H_a(2\tau))_z(v)\Big), \qquad 0 \le \tau \le \tfrac{1}{2}.$$

Continue with the path

$$K^h_t(u, v) = \Big(u, h_{a+tu}(z+v) - h_a(z)\Big), \qquad t = 2\tau - 1, \quad \tfrac{1}{2} \le \tau \le 1. \tag{C.2}$$

At the join $t = 0$, the second segment is the direct sum of I_r and Δh_a. At $t = 1$, it is $\Delta\widehat{h}$. Each K^h_t is an invertible germ: after recovering u, its second coordinate is inverted by

$$v = h^{-1}_{a+tu}(h_a(z) + w) - z.$$

The maps and their inverses depend continuously on all parameters and are the identity on a common boundary collar. After rounding the product disc, this defines $\mathscr{P}_r : \Omega^r \mathscr{F}_m \to \mathscr{F}_{m+r}$. Forgetting the framing gives the commutative diagram

$$\begin{array}{ccc} \Omega^r \mathscr{F}_m & \xrightarrow{\mathscr{P}_r} & \mathscr{F}_{m+r} \\ \downarrow & & \downarrow \\ \Omega^r \mathscr{H}_m & \xrightarrow[(h_a)\mapsto\widehat{h}]{} & \mathscr{H}_{m+r}. \end{array} \tag{C.3}$$

Here Ω^r is represented by cubical families equal to the identity near the boundary, with a fixed rounding; this convention is weakly equivalent to the usual based loop space. The embedding (4.18) identifies the rounded product with the embedded disc containing the support of the sphere diffeomorphism. These constructions and their inverses preserve a common collar for each compact simplex of parameters.

C.2. **Comparison with a smooth Gromoll representative.** For a smooth family f_a, the preferred path for a single map is the difference-quotient path from its derivative to its induced germ [16, Lemma 8.1]. Put $F(a,z)=(a,f_a(z))$. For $s,t\in[0,1]$, define

$$\mathcal{H}_{s,t}(u,v)=\begin{cases}\left(u,\dfrac{f_{a+stu}(z+tv)-f_a(z)}{t}\right), & t>0,\\ \Big(u,s(\partial_a f_a)_z u+(D_z f_a)v\Big), & t=0.\end{cases}\tag{C.4}$$

For $t>0$, the second component equals

$$\int_0^1 [s(\partial_a f)_{a+\theta stu,\,z+\theta tv}u+(D_z f)_{a+\theta stu,\,z+\theta tv}v]\,d\theta.$$

This expression is smooth in all variables down to $t=0$, where it has the value prescribed in (C.4). For $t>0$, its inverse in the second coordinate is

$$v=\frac{f_{a+stu}^{-1}(f_a(z)+tw)-z}{t};$$

for $t=0$, it is $(D_z f_a)^{-1}(w-s(\partial_a f_a)_z u)$. For every $t\in[0,1]$, the derivative of $\mathcal{H}_{s,t}$ at $(u,v)=(0,0)$ has diagonal blocks $I_r, D_z f_a$ and lower-left block $s\partial_a f_a$, and is therefore invertible. The parameter-dependent inverse function theorem supplies a smooth family of inverse germs, including at $t=0$, on uniform neighborhoods for compact parameter sets. Its restriction to $t>0$ is the displayed inverse. For fixed s, follow $t\mapsto\mathcal{H}_{s,t}$ by the path $\sigma\mapsto\mathcal{H}_{s+(1-s)\sigma,1}$, $0\le\sigma\le1$. This gives

$$\Gamma_s(\tau)=\begin{cases}\mathcal{H}_{s,2\tau}, & 0\le\tau\le\frac12,\\ \mathcal{H}_{s+(1-s)(2\tau-1),1}, & \frac12\le\tau\le1.\end{cases}\tag{C.5}$$

At $\tau=1/2$ both formulas give $\mathcal{H}_{s,1}$, and

$$\Gamma_s(0)=\begin{pmatrix}I_r & 0\\ s\partial_a f_a & D_z f_a\end{pmatrix},\qquad \Gamma_s(1)=\mathcal{H}_{1,1}=\Delta F\quad\text{for every } s.\tag{C.6}$$

At $s=1$, this is the preferred path for F, followed by a constant segment. At $s=0$, it is the difference-quotient path for f_a with a fixed, followed by (C.2). All paths fix the product boundary collar. Reparameterizing both segments by a smooth function constant near its endpoints makes the concatenation smooth and preserves its endpoints. Thus Γ_s is a homotopy over F between its associated framed homeomorphism and $\mathscr{P}_r$ applied to the framed family of f_a.

Lemma C.1 (Compatibility with stabilization)**.** *Let $m=6$ and $r\in\{1,2\}$. For a smooth r-parameter family in $\mathrm{Diff}_\partial(D^m)$ that is the identity near the entire product boundary, the smoothing comparison of* [16, Sections 8–9] *sends its Gromoll representative to the stabilization of the family comparison obtained by adding the identity on $\mathbb{R}^r$.*

Proof. In (C.3), the homotopy (C.5) identifies $\mathscr{P}_r$ on smooth families with the framed homeomorphism associated with the Gromoll representative. Over the identity family, $h_a=\mathrm{id}$ for all a and $K_t^h=\mathrm{id}$ for every t. Thus $\mathscr{P}_r$ acts on the fiber by

$$(A_{a,z},H_{a,z})\longmapsto(I_r\oplus A_{a,z},\ I_r\oplus H_{a,z}),$$

up to the constant second half of the path. This takes the direct sum of the linear and microbundle isomorphisms with the identity on $\mathbb{R}^r$. After adjunction of the parameter

and disc variables it is the stabilization map of the smoothing homotopy fibers. The spaces $\mathscr{H}_{m+r}$ and $\Omega^r\mathscr{H}_m$ at the bottom of (C.3) are contractible: the relative Alexander contraction contracts $\mathscr{H}_j$ to the identity, and its pointwise application contracts the loop space as well. Use the homotopy-fibration models of [16, Sections 8–9], replacing a projection by its mapping-path fibration if necessary. The long exact sequence of a homotopy fibration over a contractible base makes its fiber inclusion an isomorphism on all homotopy groups and on path components. Equivalently, its realization is a weak equivalence; the fiber inclusion used here is the map J in the proof of Theorem 9.1 of that paper. Applying this to the two projections in the square and using the calculated map on their fibers proves compatibility with stabilization.

By the fiber sequence used in lemma 4.6, the natural PL/PD-to-TOP comparison induces isomorphisms on π_8 and π_9. Its naturality gives the same compatibility in the PL model. After stabilization, the resulting element of $\pi_j(PL/O)$ is the smoothing class of the sphere obtained by gluing, under the isomorphism $\Psi_j : \Theta_j \to \pi_j(PL/O)$ of [4, Lemmas 2.5 and 2.7, Proposition 2.10]; see also [21, Section 2.4]. □

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